

1. REFLECTION OF LIGHT AT CURVED SURFACES

SUMMARY

1. REFLECTION OF LIGHT BY DIFFERENT SURFACES

Light travels in a straight line in homogenous medium

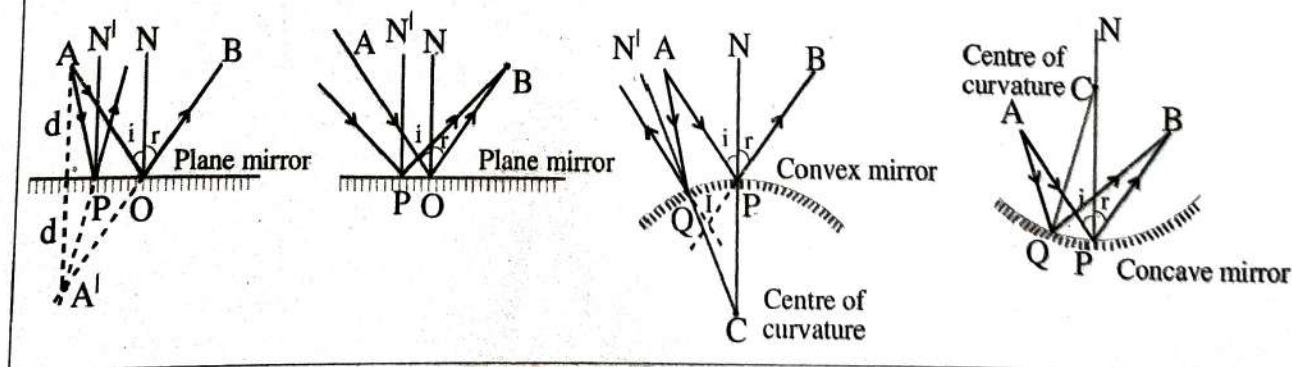
Reflection of light

- The property of light, coming back into the same medium from a boundary separating two media
- The angle of incidence is equal to the angle of reflection
- The incident ray, the normal, reflected ray, all lie in the same plane.
- To deal the reflection of light ray we require normal at point of incidence
- For a curved surface radius of curvature at a point of incidence will become normal.
- To get the radius of curvature we need to locate centre of curvature

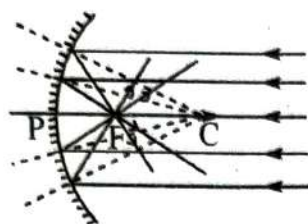
2. **Image :** The point of convergence of reflected rays from a surface is called image.

Real Image : If reflected rays are convergent we get real image. We need a screen to view it.

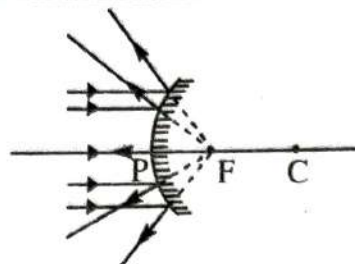
Virtual Image : If extended reflected rays are convergent we get virtual image. We can view it inside the reflecting surface .



3. Focal length of a spherical mirror :



Concave mirror



Convex mirror

When parallel beam of light is incident on a reflecting surface then the point from which the reflected rays appear to be convergent or divergent is called principal focus (F).

The distance between pole and focus is called focal length (f).

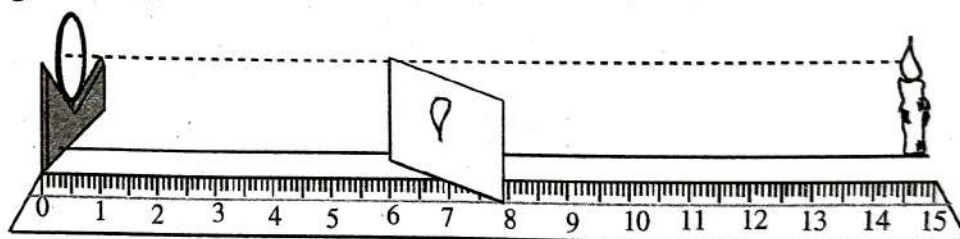
$$\text{Focal length} = \frac{\text{Radius of curvature}}{2}$$

We can determine focal length of a concave mirror by

- Forming image of the distant object : distance between pole and image formed is focal length.
- Focusing sunlight : Distance between pole and point of convergent of sunlight is focal length.

4. Observing the types of image and measuring the object distance and image distance from the mirror :

When we conduct the experiment with concave mirror of focal length (f) we can have the following inferences.

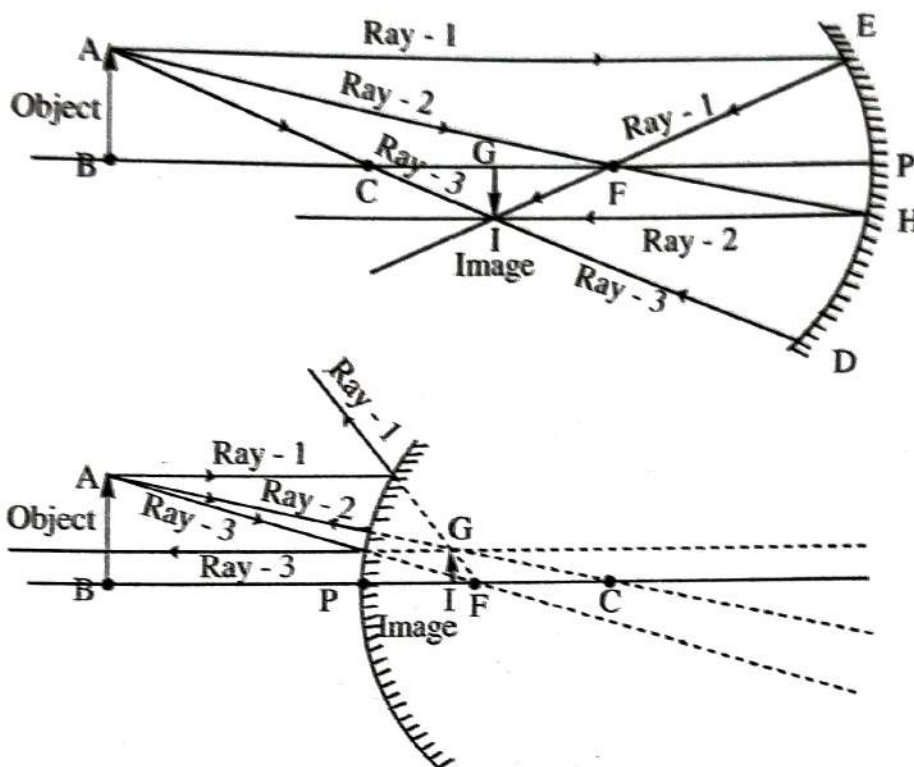


Position of the object	Position of the image	Size of the image	Nature of the image
i) At infinity ($u = \infty$)	At the focus ($v = f$)	Highly diminished, point sized	Real and inverted
ii) Beyond C ($u > 2f$)	Between F and C ($f < v < 2f$)	Diminished	Real and inverted
iii) At C ($u = 2f$)	At C ($v = 2f$)	Same size	Real and inverted
iv) Between F and C ($f < u < 2f$)	Beyond C ($v > 2f$)	Enlarged	Real and inverted
v) At F ($u = f$)	At infinity ($v = \infty$)	Highly Enlarged	Real and inverted
vi) Between P and F ($u < f$)	Behind the mirror	Enlarged	Virtual and erect

5. **Drawing Ray diagrams for the image formation due to curved surfaces :**

Although countless rays start from a given point of an illuminated object, yet it is possible to locate the position of an image, formed in a spherical mirror by taking any of the two rays discussed below.

Consider the object placed in front of the curved surface.



Ray 1 : A ray parallel to principal axis after reflection passes through the principal focus (F) or appears to be passing through principal focus.

Ray 2 : A ray passing through the principal focus (F) or appears to be passing through principal focus (F) is reflected parallel to the principal axis.

Ray 3 : A ray through the centre of curvature (C) or appears to be passing through (C) which strikes the mirror normally is reflected back along the same path.

6. **Measuring the parameters of image formation due to spherical mirrors :**

Sign convention should be applied for the use of mirror equations :

Consider coordinate axis taking the pole as origin.

All distances should be measured from the pole, denote the distances as +ve or -ve according to coordinate axis.

Magnification : It is the ratio of height of the image to the height of the object

$$m = \frac{\text{height of image}}{\text{height of object}} \Rightarrow m = \frac{-v}{u}$$

7. Mirror's formula :

Mirror's formula : Consider a point object is placed at the point O of the principal axis of a concave mirror as shown. $PO = u$, $PI = v$, $PF = f$, $PC = R$. Now we can have

Method 1 :

$$1) \beta = \alpha + \theta$$

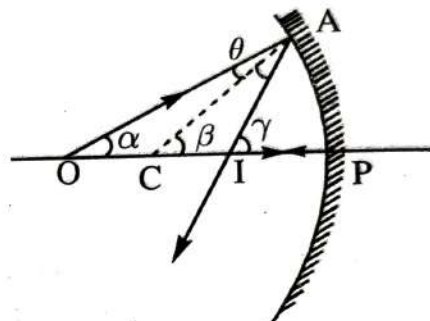
$$4) \frac{1}{PO} + \frac{1}{PI} = \frac{2}{PC}$$

$$2) \gamma = \alpha + 2\theta$$

$$3) \alpha = 2\beta - \gamma$$

$$5) \frac{1}{u} + \frac{1}{v} = \frac{1}{f} = \frac{2}{R}$$

$$\Rightarrow \alpha + \gamma = 2\beta$$



Method 2 : Consider an object AB beyond the center of curvature of a concave mirror on the principal axis as represented below in the ray diagram we get real inverted diminished image $A'B'$ formed between F and C.

We can have

$$1) \Delta ABC \sim \Delta A'B'C$$

$$\frac{AB}{A'B'} = \frac{AC}{A'C} = \frac{u-2f}{2f-v}$$

$$2) \Delta A'B'F \sim \Delta FXP'$$

$$\frac{FP'}{A'F} = \frac{XP'}{A'B'} = \frac{AB}{A'B'} = \frac{f}{v-f}$$

3) From (1) & (2) we can have

$$\frac{u-2f}{2f-v} = \frac{f}{v-f} \Rightarrow \frac{1}{u} + \frac{1}{v} = \frac{1}{f} \text{ (Mirror's formula)}$$

4) If we consider $AF = P =$ Distance of object from focus

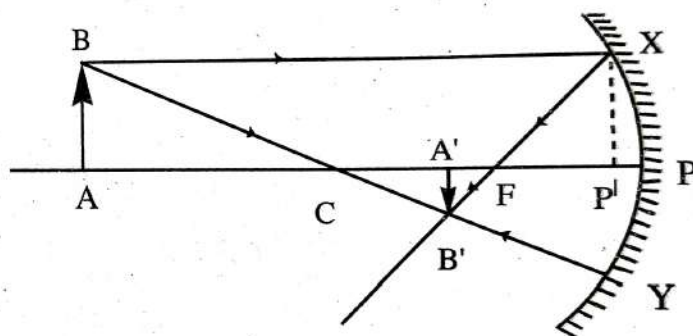
then distance of the object $u = P + f$, image distance $v = \frac{-(p+f)f}{p}$

$$m = \frac{\text{height of image}}{\text{height of object}} \Rightarrow m = \frac{-v}{u} = \frac{f}{P}$$

5) If we consider $A'F = P =$ Distance of the image from focus

then distance of the image $v = P + f$, then distance of the object $u = \frac{-(p+f)f}{p}$

$$m = \frac{\text{height of image}}{\text{height of object}} \Rightarrow m = \frac{-v}{u} = \frac{P}{f}$$



$$AP = u$$

$$PA' = v$$

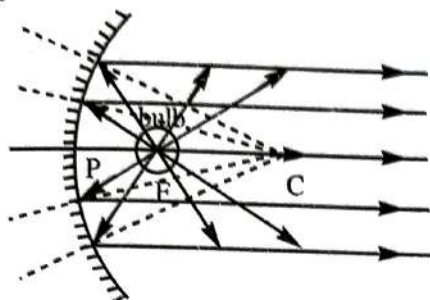
$$PC = 2f$$

$$PA' = f$$

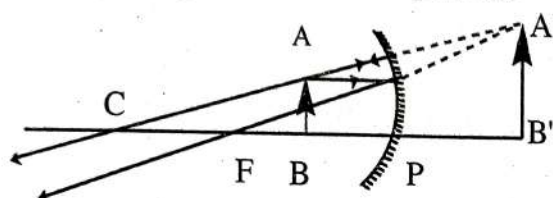
8.

Uses of Concave Mirrors

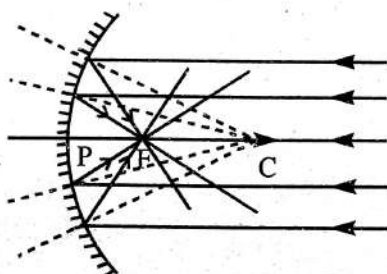
1) Used in torch lights to get parallel beam of light



2) These are used as shaving mirrors to see a larger image of the face, Dentists use this to see large images of the teeth of patients.

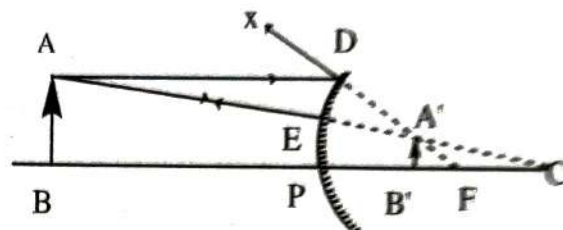


3) Used to concentrate sunlight to produce heat in solar furnaces.

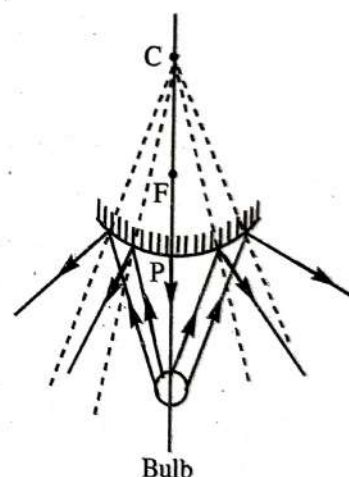


Uses of Convex Mirrors

1) Used as rear view mirrors in vehicles.



2) Used in diffusing the light from the bulbs in the flood lights



PRIORITY - I

Conceptual understanding (AS₁)

- The magnification produced by a plane mirror is + 1. What does this mean? (Part- I.Pg.No.31)
- State the laws of reflection of light. (Part- I. Pg.No.18)
- Distinguish between real and virtual image ? (S.S.C - 2019) (Part- I. Pg.No.28)
- How do you find the focal length of a concave mirror ? (S.S.C - 2017) (Part- I. Pg.No.38)
- State the differences between convex and concave mirrors. (T.Q)(Part- I. Pg.No.28)
- What do you know about the terms given below related to the spherical mirrors ? (T.Q) (Part- I. Pg.No.29)

a) Pole	b) Centre of curvature	c) Focus	d) Radius of curvature
e) Focal length	f) Principal axis	g) Object distance	h) Image distance
i) Magnification			

7. Write the rules mentioned for sign convention. (Part- I. Pg.No.41)
8. Why do we prefer a convex mirror as a rearview mirror in vehicles? (Part- I. Pg.No.33)
9. Where will the image form when we place an object, on the principal axis of a concave mirror at a point between focus and centre of curvature ? (Part- I. Pg.No.28)
10. The magnification of a concave mirror is equal to -1 then, (S.S.C - 2018) (C.C.E)
 - a) Find the position of object and image. (Part- I. Pg.No.45)
 - b) Discuss the nature of the image formed ?

Asking questions and making hypothesis (AS₂)

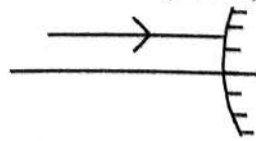
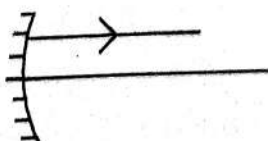
1. By observing steel vessels and different images in them; Surya a third class student asked some questions his elder sister Vidya. What may be those questions ? (Part- I. Pg.No.41)
2. Imagine that spherical mirrors are not known to human being, guess the consequences. (Part- I. Pg.No.31)

Experimental skills and field investigation (AS₃)

1. What do you infer from the experiment which you did with concave mirror and measured the distance of object and distance of image ? (T.Q) (Part- I. Pg.No.30)
2. Find the nature and position of images, when an object is placed at different places on the principal axis of a concave mirror. (S.S.C - 2016, 2018) (T.Q) (Part- I. Pg.No.34)
3. List out the material required and write the experimental procedure to find the image distance using concave mirror when an object is kept in front of it on its principal axis at different locations. (S.S.C - 2023) (C.C.E) (Part- I. Pg.No.46)

Communication through drawing, diagrams and graphs (AS₅)

1. Draw suitable rays by which we can guess the position of the image formed by a concave mirror. (T.Q) (Part- I. Pg.No.32)
2. Draw a ray diagram in each of the following cases to show the position nature of image formed when the object is placed:
 - i) Between pole and focus of a concave mirror. (TextBook Pg.No.7,8,9)
 - ii) Between focus and centre of curvature of a concave mirror.
 - iii) at the centre of curvature of a concave mirror
 - iv) between infinity and pole of a convex mirror
 - v) at infinity from a convex mirror.
3. To form the image on the object itself, how should we place the object in front of a concave mirror. Explain with a ray diagram. (T.Q) (Part - I.Pg.No. 33)
4. Complete the following ray diagrams and locate the focal point, centre of curvature and pole of the given mirror. (C.C.E) (Part- I. Pg.No.49)



Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

1. How do you appreciate the role of spherical mirrors in our daily life ? (S.S.C - 2019)
(Part - I. Pg.No. 44)
2. Write the uses of convex and concave mirrors. (S.S.C - 2022)
(Part - I. Pg.No. 42)

PRIORITY - II

Conceptual understanding (AS₁)

1. How do you get a virtual image using a concave mirror ? (T.Q) (Part- I. Pg.No.29)
2. Define magnification and write the expression for magnification ? (Part- I. Pg.No.40)
3. Find the distance of image, when an object is placed on the central axis, at 10 cm distance in front of a concave mirror whose radius of curvature is 8 cm. (T.Q) (Part- I. Pg.No.31)
4. The focal length of a convex mirror is 50 cm then calculate its radius of curvature ? (C.C.E)
(Part- I. Pg.No.46)
5. An object is placed at the following distances from a concave mirror of focal length 10 cm.
a) 8 cm. b) 15 cm. c) 20 cm. d) 25 cm. (C.C.E)
Which position of the object will produce..... (Part- I. Pg.No.46)
i) a diminished, real and inverted image ? ii) a magnified, real and inverted image ?
iii) a magnified, virtual and erect image ? iv) an image of same size as the object ?
Justify your answer in each case.
6. 2cm height object is placed at 12 cm distance from the pole of concave mirror and the image formed at 4 cm from the pole. Find the focal length of the mirror and height of the image. (C.C.E)
(T.Q) (Part- I. Pg.No.46)
7. A convex mirror with a radius of curvature 3m is used as rearview in automobile. If a bus is located at 5m from the mirror, find the position, nature and size of the image ? (T.Q) (Part - I. Pg.No. 33)

Asking questions and making hypothesis (AS₂)

1. A teacher asked the students, "which mirrors can form virtual images?" Saketh replied, "Concave mirrors"; Spandana replied, "Plane mirrors"; Reeta replied "Convex mirrors". What is your opinion ? Which answer do you support ? Explain.
(Part- I. Pg.No.47)(C.C.E)

Communication through drawing, diagram and graphs (AS₅)

1. Show the formation of image with a ray diagram, when an object is placed on the principal axis of a concave mirror away from the centre of curvature.
(Part- I. Pg.No.32)
2. Explain the process of making Solar cooker.
(Part- I. Pg.No.43)

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

1. How do you appreciate reflection of light by concave mirror in making of TV antenna dishes ?
(Part- I. Pg.No.44)
2. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.
(Part- I. Pg.No.45)

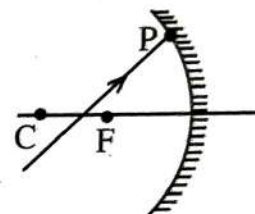
PRIORITY - III

Conceptual understanding (AS_1)

1. The focal length of a concave mirror in air is f . When it is completely immersed in water will there be any change in its focal length or not? Why? (C.C.E) (T.Q) (Part- I. Pg.No.47)
2. How parabolic mirror is better than concave mirror in collecting solar energy? (Part- I. Pg.No.40)

Communication through drawing, diagram and graphs (AS_2)

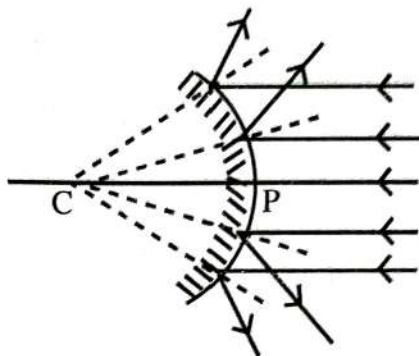
1. a) What type of mirror is used in the car headlights? (C.C.E) (Part- I. Pg.No.48)
 b) Where do the bulb is placed with respect to the mirror in the head lights of a car?
 c) Draw the ray diagram representing the light rays emitted from bulb and their reflection after incident on the mirror?
2. A light ray is incident on a concave mirror as shown in the above diagram and getting reflected from the mirror. Then answer the following questions. (C.C.E) (Part- I. Pg.No.48)
 a) What happens to the light rays travelling parallel to principal axis when they are incident on the concave mirror?
 b) What happens to the light ray which is passing through the principal focus after incident on the concave mirror?
 c) Basing on the answers given to the above questions draw the reflected ray for the ray incident at P in the above diagram?



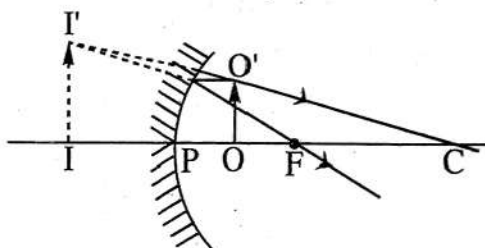
SESSIONWISE STUDENT TASK

1. Predict and write the reason, why the value of the distance of the object is always negative in the mirror equation.
2. Draw the ray diagram of image formed by the rear view mirror in vehicles?
3. Draw suitable ray diagrams and write nature of image when an object is placed on the principal axis at a distances of 3 cm., 6 cm. in front of a concave mirror whose focal length is 4 cm.
4. What happens if concave mirrors are used as rear view mirrors of vehicles?
5. An object is placed in front of concave mirror whose magnification is +1.75. What does it mean?
6. What changes may happen to the images formed by a concave mirror when an object is moving from beyond 'C' to towards pole of the mirror on its principal axis?
7. Ekambaram is doing an experiment using mirrors to understand the concept of magnification. He identified the situation where the magnification is found to be greater than one.
 a) What type of mirror is he used in this experiment?
 b) What is type, position of the object for the above situation?

8. Sukumar watched his face in the side view mirror of a car and noticed that his image appearing to be smaller than his face. Then answer the following
 - a) What kind of mirror it is ?
 - b) What is the nature of the image observed by sukumar ?
 - c) Draw the ray diagram indicating image formation ?
9. See the given figure. A set of parallel rays are falling on a convex mirror. What conclusions can you draw from this ?



10. What type of the image is formed in the below diagram ? Explain the characteristics of image



11. The teacher asked James a question. James replied the correct answer " $R = 2f$ ". Can you guess, what is the question ?
12. While doing an experiment with concave mirror, Anand identified size of an image equal to the size of object.
 - a) Where the object is placed ?
 - b) What is the magnification value at this position ?
 - c) Draw the ray diagram for it ?
13. Frame some questions to know about the Focus and centre of curvature of spherical mirror ?
14. Write any two differences between the image formed by a concave mirror when the object is placed beyond 'C' and image formed when object is placed in front of the a convex mirror.
15. Draw a suitable ray diagram of image formed by a mirror, in which object distance is negative, image distance is positive, magnification is positive and less than 1.
16. What type of mirrors are used in vehicle head lamps ? Write any two uses of them in daily life ?

17. Observe the following table and answer the following :

Type	Characteristic property	Image size
Concave mirror	Real image	enlarged
Convex mirror	Virtual image	diminished

- 1) Which mirror is used as rear view mirror in vehicles ?
- 2) Which mirror is used in car head lights ?

18. The information about the image formed in concave mirror is given in the following table.

S. No.	Object	Image	Magnification
1)	X	Erected	+ 2.75
2)	Y	Inverted	- 2.75
3)	Z	Same size that of object	- 1.0

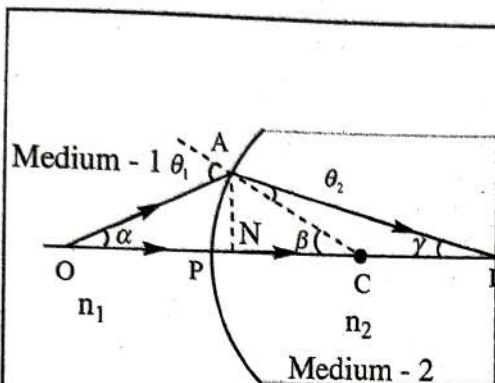
Answer the following questions.

- i) Which object is forming virtual image ?
- ii) Name the object which is placed at centre of the curvature ?
- iii) Name the object which formed erected image.
- iv) If the size of the object 'Z' is 3 cm, then what is the size of the image ?

2. REFRACTION OF LIGHT AT CURVED SURFACES

SUMMARY

1. When light rays close to principal axis are incident on curved surface separating two different media.



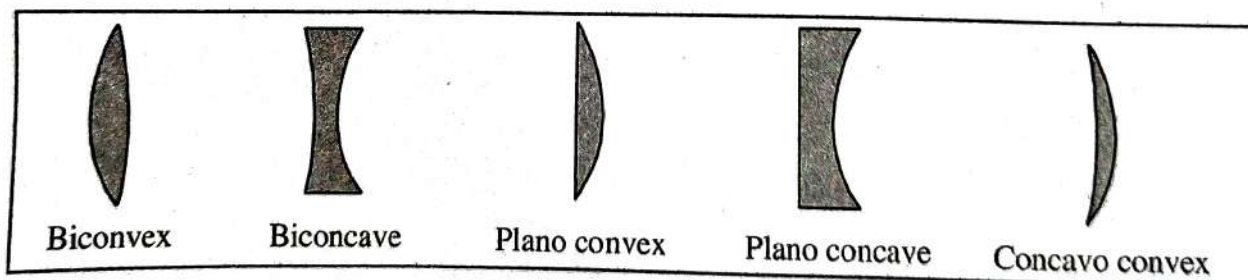
- We need to consider normal at point of incidence
- To plot the normal we require centre of curvature (C) of the curved surface.
- We have to apply snell's law of refraction to get the distance of image (v) inside the second medium for the object placed at a distance (u) in the first medium.
- $n_1 \sin \theta_1 = n_2 \sin \theta_2$ [$\because \theta_1, \theta_2$ are very small]
 $\Rightarrow n_1 \theta_1 = n_2 \theta_2$
- We get for curved surface $\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$
- In the case of a plane surface, radius of curvature (R) approaches infinity. Hence the formula for plane surface will be

$$\frac{n_2}{v} - \frac{n_1}{u} = 0 \rightarrow \frac{n_2}{v} = \frac{n_1}{u}$$

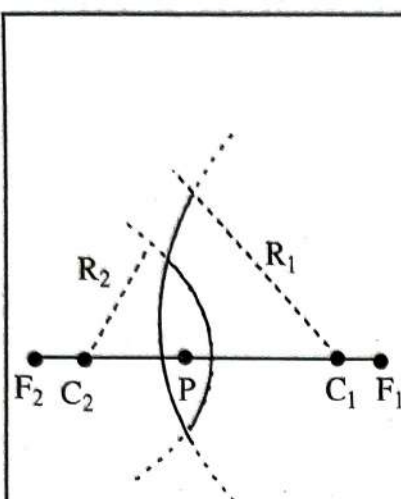
2.

Lenses : A lens is formed when a transparent material is bounded by atleast one curved surface or both. Lenses can be of various types. Some typical lenses along with their names are shown.

For drawing ray diagrams related to lenses we represent convex lens with a symbol $\updownarrow$ and concave lens with $\frown$



3. Important terms of the lens :



F_1, C_1 and F_2, C_2
may or may not
coincide

Centre of curvature : Each curved surface of a lens is part of a sphere. The centre of the sphere which contains the part of the curved surface is called centre of curvature.

Principal axis : The line joining the two centres of curvature is called principal axis.

Optic centre : The midpoint of a thin lens is called optic centre of lens (P).

Focal point or focus : The point (F) at which a parallel beam of light incident on a lens converges to a point or seems to emanate from a point on the principal axis is called focal point or focus (F).

Focal length of the lens : Every lens has two focal points. The distance between the focal point and optic centre is called the focal length of lens denoted by 'f'.

4. Observing the types of image and measuring the object distance and image distance from the convex lens :

When we conduct the experiment with convex lens of focal length (f) we can have the following inferences.

i) We may not get any real image at some positions, those positions are when object is placed at F.

ii) We can establish the relationship between u, v and f as $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

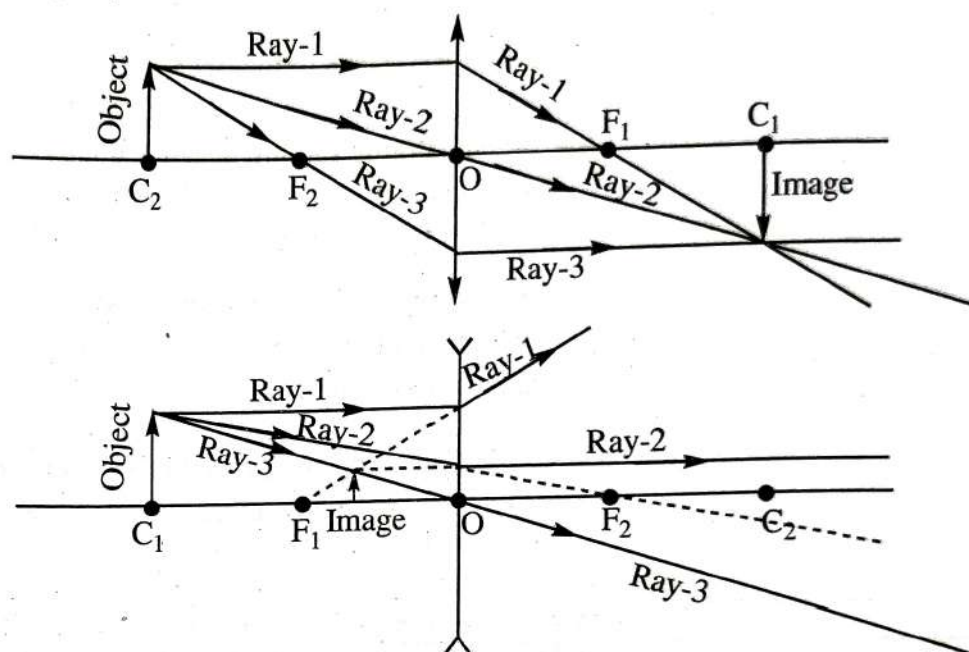
Position of the object	Position of the object	Size of the image	Nature of the image
i) At infinity $u = \infty$	At the focus $v = f$	point sized	Real
ii) Beyond C_2 $u > 2f$	Between F_1 and C_1 $f < v < 2f$	Diminished	Real and inverted
iii) At C_2 $u = 2f$	At C_1 $v = 2f$	Same size	Real and inverted
iv) Between C_2 and F_2 $f < u < 2f$	Beyond C_1 $v > 2f$	Enlarged	Real and inverted
v) At F_2 $u = f$	At infinity $v = \infty$	Can't say	Can't say
vi) Between O and F_2 $u < f$	Behind the lens	Enlarged	Virtual and erect

5.

Drawing Ray diagrams for the image formation due to the spherical lenses :

Although countless rays start from a given point of an illuminated object, yet it is possible to locate the position of an image, formed in a spherical lens by taking any of the two rays discussed below.

Consider the object placed in front of the convex lens.



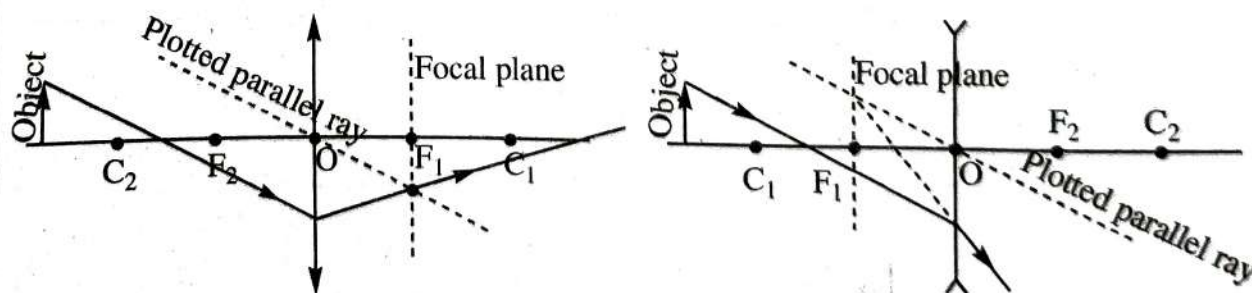
Ray 1 : We know that the rays passing parallel to the principal axis converge at the focus or appear to diverge from the focus.

Ray 2 : Light rays passing through the focus or appear to travel through focus will take a path parallel to principal axis after refraction.

Ray 3 : Any ray passing through the optic centre will travel undeviated.

To track the path of the oblique rays from the object after refraction through the lens

1. Plot a ray parallel to incident ray through optic centre.
2. Draw the refracted ray from the point of incidence passing through the point of intersection of focal plane and plotted parallel ray as shown below.



6. Determining focal length of a given lens :

Focal length of a given lens depends on

Radii of curvatures R_1, R_2

Refractive index of the material of the lens n_2

Refractive index of the medium surrounding the lens n_1

Power of the lens (P)

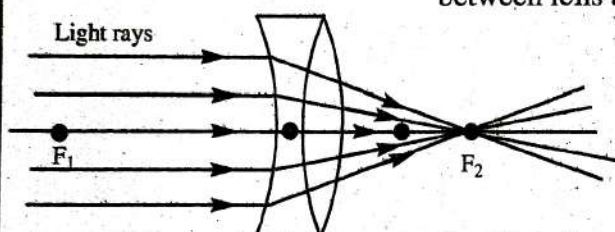
The reciprocal of focal length taken in meters is called power of the lens. $P = \frac{1}{f}$

When convex lens is given determination of focal length

The maximum limiting distance upto which we can observe virtual image of an object placed in front of the convex lens is called focal length.

The minimum limiting distance upto which we can observe real image of an object placed in front of the convex lens is called focal length.

While focussing sunrays using a convex lens the distance between lens and focussing point is taken as focal length.



When concave lens is given determination of focal length

Combine the given concave lens of unknown focal length (x) with a convex lens known focal length (y) such that the total combination act as converging lens.

Measure the focal length of the combination as f by focussing the sunrays as shown.

Using the formula power of the lens combination $P = P_1 + P_2$
 $\Rightarrow \frac{1}{f} = \frac{1}{x} + \frac{1}{y}$ We can determine x value.

7. Measuring the parameters of image formation due to lenses :

Sign convention should be applied for the use of lens formula :

Consider coordinate axis taking the optic centre as origin.

All distances should be measured from the optic centre, denote the distances as +ve or -ve according to coordinate axis.

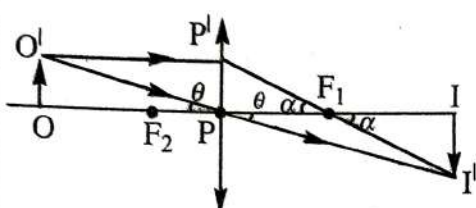
Magnification : It is the ratio of height of the image to the height of the object

$$m = \frac{\text{height of image}}{\text{height of object}} \Rightarrow m = \frac{v}{u}$$

Lens formula :

Observe the following diagram depicting the formation of image for an object placed in front of a convex lens, taking $OP = u$, $PI = v$, $PF_1 = PF_2 = f$.

We can have



$$1. \triangle OO'P \sim \triangle PII' \Rightarrow \frac{OO'}{II'} = \frac{PO}{PI} = \frac{u}{v}$$

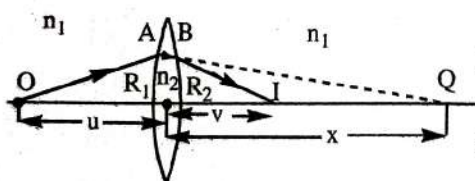
$$2. \triangle PP'F_1 \sim \triangle F_1II' \Rightarrow \frac{PP'}{II'} = \frac{OO'}{II'} = \frac{PF_1}{PI - PF_1} = \frac{f}{v - f}$$

$$\text{From (1) \& (2) we can have } \frac{u}{v} = \frac{f}{v - f} \Rightarrow \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Using sign convention the lens formula is denoted as $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Lens makers formula :

Consider a ray, from 'O' which is incident on the convex surface of the lens with radius of curvature R_1 at A as shown in figure and Q is the image of the object due to first surface I is the image due to second surface $n_1 = n_a$ (surrounding medium), $n_2 = n_b$ (lens material) then we can have



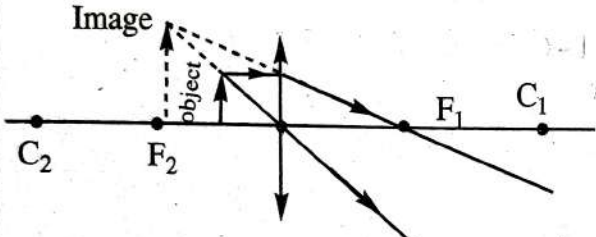
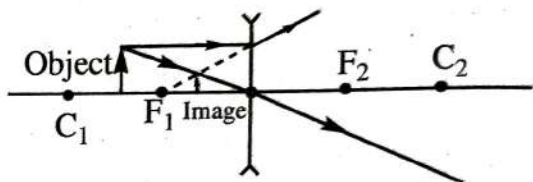
$$1. \text{ At the first surface } \frac{n_b}{x} + \frac{n_a}{u} = \frac{(n_b - n_a)}{R_1}$$

$$2. \text{ At the second surface } \frac{n_a}{v} - \frac{n_b}{x} = \frac{(n_b - n_a)}{R_2}$$

$$3. \text{ Solving (1) \& (2) we can have } \frac{1}{v} + \frac{1}{u} = \left(\frac{n_b}{n_a} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\text{Using sign convention we can have } \frac{1}{v} - \frac{1}{u} = \left(\frac{n_b}{n_a} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{f}$$

8. Applications of convex and concave lenses :

Uses of Convex Lenses :	Uses of Concave Lenses :
i) Convex lenses are used in spectacles to correct the defect of vision called hypermetropia (or long sightedness). ii) Convex lens is used for making a simple camera. iii) Convex lens is used as a magnifying glass (or magnifying lens)(by palmists, watchmakers, etc.). iv) Convex lenses are used in making microscopes, telescopes and slide projectors (or film projection).	i) Concave lenses are used in spectacles to correct the defect of vision called myopia (or short - sightedness). ii) Concave lens is used as eye - lens in Galilean telescope. iii) Concave lens is used in wide - angle spyhole in doors.
	

PRIORITY - I

Conceptual understanding (AS₁)

- What do you know about the terms given below related to the spherical lenses ?
(C.C.E) (Part - I . Pg.No. 100)

a) Optic centre	b) Centre of curvature	c) Focus	d) Radius of curvature
e) Focal length	f) Principal axis	g) Object distance	h) Image distance
- Write the lens maker's formula and explain the terms in it. (Part - I . Pg.No. 98)
- Can virtual image be photographed by camera? (T.Q) (Part - I. Pg.No. 99)
- What are the steps involved in sign convention. (Part - I.Pg.No. 96)
- Write the rules to draw ray diagrams for image formation by lenses ? (S.S.C - 2016)
(Part - I. Pg.No. 97)
- An object is placed at the following distances from a convex lens of focal length 10 cm.
(C.C.E) (T.Q) (Part - I . Pg.No. 101)

a) 8 cm.	b) 15 cm.	c) 20 cm.	d) 25 cm.
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Which position of the object will produce (S.S.C - 2015) .

i) a diminished, real and inverted image ?	ii) a magnified, real and inverted image ?
iii) a magnified, virtual and erect image ?	iv) an image of same size as the object ?

Justify your answer in each case.

Asking questions and making hypothesis (AS₂)

1. Harsha tells Siddhu that the double convex lens behaves like a convergent lens. But Siddhu knows that Harsha's assertion is wrong and corrected through by asking some questions what are the questions asked by Siddhu? (Part- I. Pg.No.99)
2. A convex lens is made up of three different materials as shown in the figure. How many no of images does it form? (S.S.C - 2016) (Part- I. Pg.No.87)



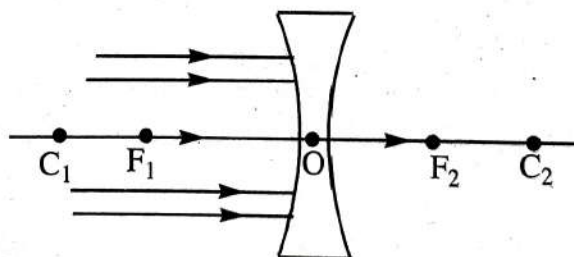
3. The refractive index of convex lens material is 1.46. The refractive index of Benzene and water is 1.5 and 1.33 respectively. How does the lens behave when it is kept in Benzene and water? Guess and write. (S.S.C - 2018) (C.C.E) (Part- I. Pg.No.102)

Experimental skills and field investigation (AS₃)

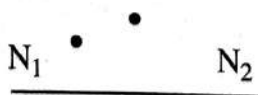
1. Write the names of apparatus used in the experiment to show that focal length of a lens depends on surrounding medium where it was placed (C.C.E) (Part- I. Pg.No.102)
2. How do you find the focal length of a lens experimentally? (T.Q) (Part- I. Pg.No.83)
3. You have a lens. Suggest an experiment to find out the focal length of the lens. (T.Q) (Part- I. Pg.No.87)
4. How do you verify experimentally that the focal length of a convex lens is increased when it is kept in water. (S.S.C - 2019) (T.Q) (Part- I. Pg.No.83)

Communication through drawing, diagrams and graphs (AS₅)

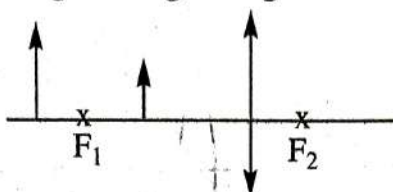
1. How do you represent convex lens and concave lens while drawing ray diagrams? (C.C.E) (Part- I. Pg.No.103)
2. Complete the following diagram (C.C.E) (Part- I. Pg.No 103)



3. Figure shows a point light source and its image produced by a lens with an optical axis N_1N_2 . Find the position of the lens and its foci using a ray diagram. (Part- I. Pg.No.88)



4. Draw the ray diagrams of image formation due to convex lens by placing the object at various positions. (Text Book Pg.No.68, 69)
5. Draw the ray diagrams of image formation due to concave lens by placing the object at various positions. (Text Book Pg.No.69)
6. Draw ray diagrams for the following positions and explain the nature and position of image for convex lens (Part- I. Pg.No.84)
 - i) Object is placed at C_2
 - ii) Object is placed between F_2 and optic centre O.
7. Write the characteristics of the image in the given figure. (C.C.E) (Part- I. Pg.No.103)



8. Write the uses of lenses in daily life. (S.S.C - 2022) (C.C.E) (Part - I. Pg.No. 102)

Aesthetic sense/ appreciation/ values (AS_6) / Application to daily life, concern to bio diversity (AS_7)

1. How do you appreciate the coincidence of the experimental facts with the results obtained by a ray diagram in terms of behaviour of images formed by lenses ? (Part- I. Pg.No.99)
2. Write the applications of convex and concave lenses in daily life with neat ray diagrams. (Part- I. Pg.No.102)

PRIORITY - II

Conceptual understanding (AS_1)

1. Two converging lenses are to be placed in the path of parallel rays so that the rays remain parallel after passing through both lenses. How should the lenses be arranged? Explain with a neat ray diagram. (Part- I. Pg.No.85)
2. The focal length of a converging lens is 20cm. An object is 60 cm from the lens where will the image be formed and what kind of image is it ? (Part- I. Pg.No.85)
3. One portion of a convex lens is covered as shown, will the lens produce a complete image of the object? Describe in brief the characteristics of the image formed (Part- I. Pg.No.93)

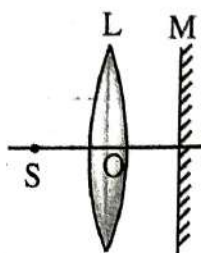


4. A double convex lens has two surfaces of equal radii ' R ' and refractive index $n = 1.5$, then find focal length ' f '. (T.Q) (Part - I. Pg.No. 86)

5. Mention the main features of Galileo Telescope. (Part - I. Pg.No. 92)
6. Find the refractive index of glass which is a symmetrical convergent lens if its focal length is equal to the radius of curvature of its surface. (T.Q) (Part- I. Pg.No.94)

Asking questions and making hypothesis (AS₂)

1. The diagram shows a point source of light S, a convex lens L and a plane mirror M. These are placed such that rays of light from S return to it after reflection from M. (C.C.E) (Pg. No. 104)
 - i) What is the distance OS called
 - ii) To which point (left of S on S or right of S) will the rays return, if M is moved to the left and brought in contact with L?



2. What changes may be observed in the image when the object moves from focus to beyond centre of curvature using a convex lens. (C.C. E) (Pg. No. 86)

Experimental skills and field investigation (AS₃)

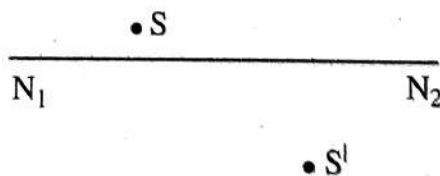
1. Let us assume a system that consists two lenses with focal length f_1 and f_2 respectively. How do you find the focal length of the system experimentally, when (T.Q) (Part- I. Pg.No.91)
 - i) Two lenses are touching each other (S.S.C - 2017)
 - ii) They are separated by a distances 'd' with common principal axis.

Information Skills. (AS₄)

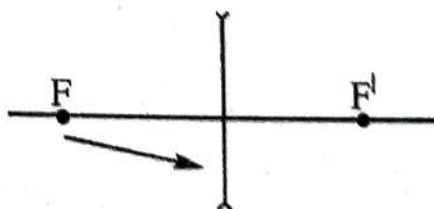
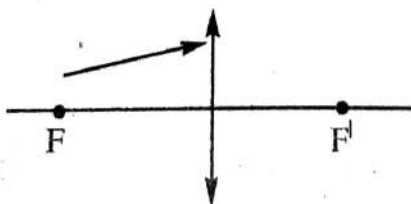
1. An electric lamp and a screen are placed on the table, in a line at a distance of 1m. In what positions of convex lens of focal length of $f = 21$ cm will the image of lamp be sharp? (Part- I. Pg.No.81)

Communication through drawing, diagram and graphs (AS₅)

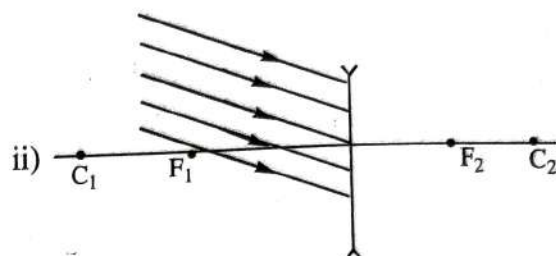
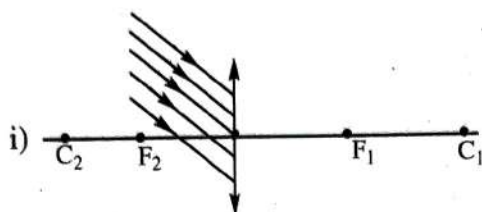
1. Find the focus by drawing a ray diagram using the position of source S and the image S' given in the figure. (T.Q) (Part- I. Pg.No.88)



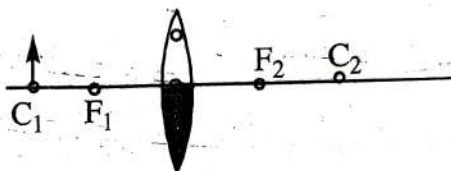
2. Complete the ray diagram to show the paths of the rays after refraction through the lenses shown in the figures (a) and (b)? (Part- I. Pg.No. 82)



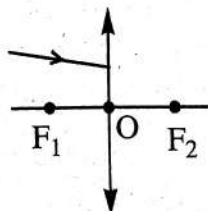
3. A parallel beam of rays is incident on a convergent lens with a focal length of 40 cm. Where should a divergent lens with a focal length of 15 cm be placed for the beam of rays to remain parallel after passing through the two lenses ? Draw a ray diagram. (T.Q) (Part- I.Pg.No. 89)
4. What happens when parallel rays of light fall on a lens making some angle with the principal axis ? Complete the following diagrams. (S.S.C - 2019) (Part- I. Pg.No.95)



5. One half of a convex lens is covered with a black paper. Will the lens produce a complete image of the object write your observations and complete the diagram. (Part- I. Pg.No. 93)



6. Complete the ray diagram. (S.S.C - 2022)



PRIORITY - III

Conceptual understanding (AS₁)

1. A double convex lens has two surfaces of equal radii R and refractive index n is taken in air then answer the following. (C.C.E) (Part - I Pg. No. 104)
- Calculate the focal length of given lens ?
 - As an object is placed at one of its centre of curvatures then what will be the distance of the image formed ?
2. The refractive index of the material of the lens is 1.5 if an object is placed at a distance of 30 cm in front of the lens then image is formed at a distance of 20 cm (C.C.E) (Part - I, Pg.No . 105)
- Calculate the focal length of the lens.
 - What kind of lens is it ?
 - If two surfaces of the lens have same radii then calculate its value ?

3. Draw the ray diagrams for the following conditions for a convex lens having focal length $f = 30\text{cm}$ (C.C.E) (Part-I, Pg.No 107)
 - a) object placed at 30 cm
 - b) object placed at 60 cm
 - c) object placed at 10 cm
 - d) object placed at 45 cm
4. What is the focal length of double concave lens kept in air with two spherical surfaces of radii $R_1 = 30\text{ cm}$ and $R_2 = 60\text{cm}$. Take refractive index of lens as $n = 1.5$ (Part- I. Pg.No. 82)

Information Skills (AS₄)

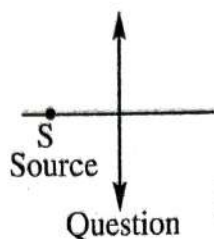
1. Draw an arrow of length 4 cm using a black sketch pen on a thick sheet of paper. Take an empty cylindrical -shaped transparent vessel such as glass tumbler. Keep it on the table. Ask your friend to bring the sheet of paper on which arrow was drawn behind the vessel while you look at it from the otherside (The arrow mark should be in horizontal position). (C.C.E) ((Part - I, Pg. No. 107)
 - a) What do you see ?
 - b) Why do you see a diminished image ?
 - c) Is the image real or virtual ?
 - d) Can you draw a ray diagram showing how it is formed ?

Ask your friend to fill the vessel with water. Look at the figure of arrow from the same position as before.

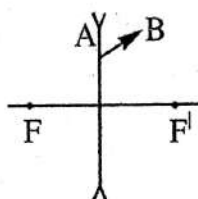
 - e) What do you see now ?
 - f) Do you get an inverted image ?
 - g) How could this happen ?
 - h) Can you draw a ray diagram showing how it is formed ?
2. You might have observed that a lemon in a glass of water appears bigger than its actual size, when viewed from the sides of tumbler. (C.C.E)(Part - I, Pg.No. 108)
 - a) How can you explain this change in the size of lemon ?
 - b) Is the lemon that appears bigger in size an image of lemon or is it the real lemon ?
 - c) Can you draw a ray diagram to explain this phenomenon ?
3. Your friend told the following formulae $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ and asked to answer the following questions. (C.C.E)(Part - I. Pg.No.109)
 - a) What are conditions required to apply the given formulae ?
 - b) What are the situations in which we can use that formulae ?
 - c) What are the precautions to be taken while using the given formulae ?

Communication through drawing, diagram and graphs (AS₃)

1. Draw a ray diagram to locate the position of image when a point source (S) is placed on optical axis N of a convex lens, in such a way that it is beyond focal point (F). See figure. (Part- I. Pg.No. 82)



2. Figure shows ray AB that has passed through a divergent lens. Construct the path of the ray up to the lens if the position of its focal point is known. (T.Q) (Part- I. Pg.No.88)



3. Ravi said that a convex lens can converge any beam of light rays even they are parallel or convergent or divergent. But Khader thought that it is not correct. Draw the ray diagrams to explain which of these two opinions are correct? (C.C.E) (Part - I. Pg.No.109)

SESSIONWISE STUDENT TASK

1. Magnification of an image formed due to a convex lens is -1.5. Guess and write the position of object and image. (S.S.C - 2022)
2. Two convex lenses of same focal length are fixed in a PVC pipe at a distance double to their focal length. What happens if a boy sees the moon with that arrangement? (S.S.C - 2016)
3. What type of image is observed if an object is placed between the convex lens and its centre of curvature?
4. What happen, if a parallel beam of light incident on a convex lens making some angle with principal axis?
5. How do you say that focal length of concave lens is always negative?
6. How light rays behave when passing through optic centre of convex lens and concave lens?
7. What happens, if concave lens is used as magnifier instead of convex lens in micro scope?
8. To get the magnified image with the microscope, where should the object be placed?
9. Is the focal length of a lens is same in all media? Predict.
10. A lemon in a glass of water is viewed from the sides of the glass appears bigger than that of actual size. Why?

11. Observe the information regarding convex lens and answer the following questions.

S.No.	Object	Magnification (m)
1	A	+3
2	B	-1
3	C	-0.6
4	D	-3

Questions:

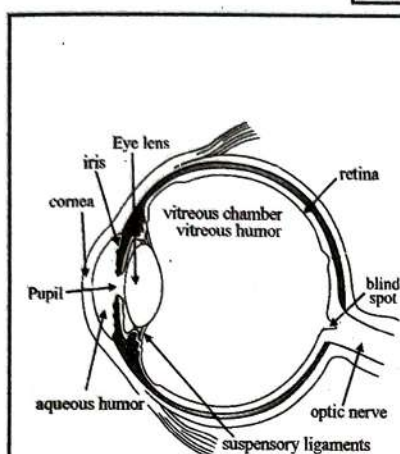
- i) Write the image characteristics for the object 'A'?
 - ii) Write the object position and image position of the object 'D'?
 - iii) Write the image characteristics of image formed by object 'B'?
 - iv) Draw the ray diagram showing image formed by object 'C'?
12. During lens experiments when ever virtual image is formed, it is not possible to measure the image distance. Is this information true or not ? Explain with reasons ?
 13. You are provided with a printed piece of paper. Using this paper how will you differentiate between a convex lens and a concave lens ?
 14. A person is viewing an extended object. if a converging lens is placed infront of his eye will he feel that the size of object is increased ? Why ?
 15. Rajesh used a lens to burn a piece of paper. What is the lens ? Draw a ray diagram related to this process.
 16. If Jagruthi obtained a magnification of -1 with convex lens of focal length 10 cm . What is the object distance from the lens ?
 17. Which lens gives always virtual image to any real object irrespective of its position. Draw the ray diagram for it.
 18. It is noticed that small air bubbles present in water are diverging the incident light. Give reasons for this.
 19. Sudhakar found focal length of a biconvex lens in laboratory.
 - a) Does he used screen or not ? Explain why ?
 - b) What are the apparatus he used for the experiment and draw the table he needs to tabulate the values.
 - c) Write the formula he used ?
 - d) What are the precautions he needs to take in this experiment to find the focal length more accurate?

3. HUMAN EYE & COLOURFUL WORLD

SUMMARY

1.

CONSTRUCTION OF HUMAN EYE



Cornea : It is a transparent protective layer in human eye

Pupil : It regulates and controls the amount of light entering eye.

Iris : It is a dark muscular diaphragm that controls the size of the pupil.

Ciliary muscles : These can alter the curvature and thickness of the eye lens.

Crystalline lens : It provides the finer adjustment of focal length

Retina : It is a light - sensitive screen of the eye on which a real and inverted image of the object is formed.

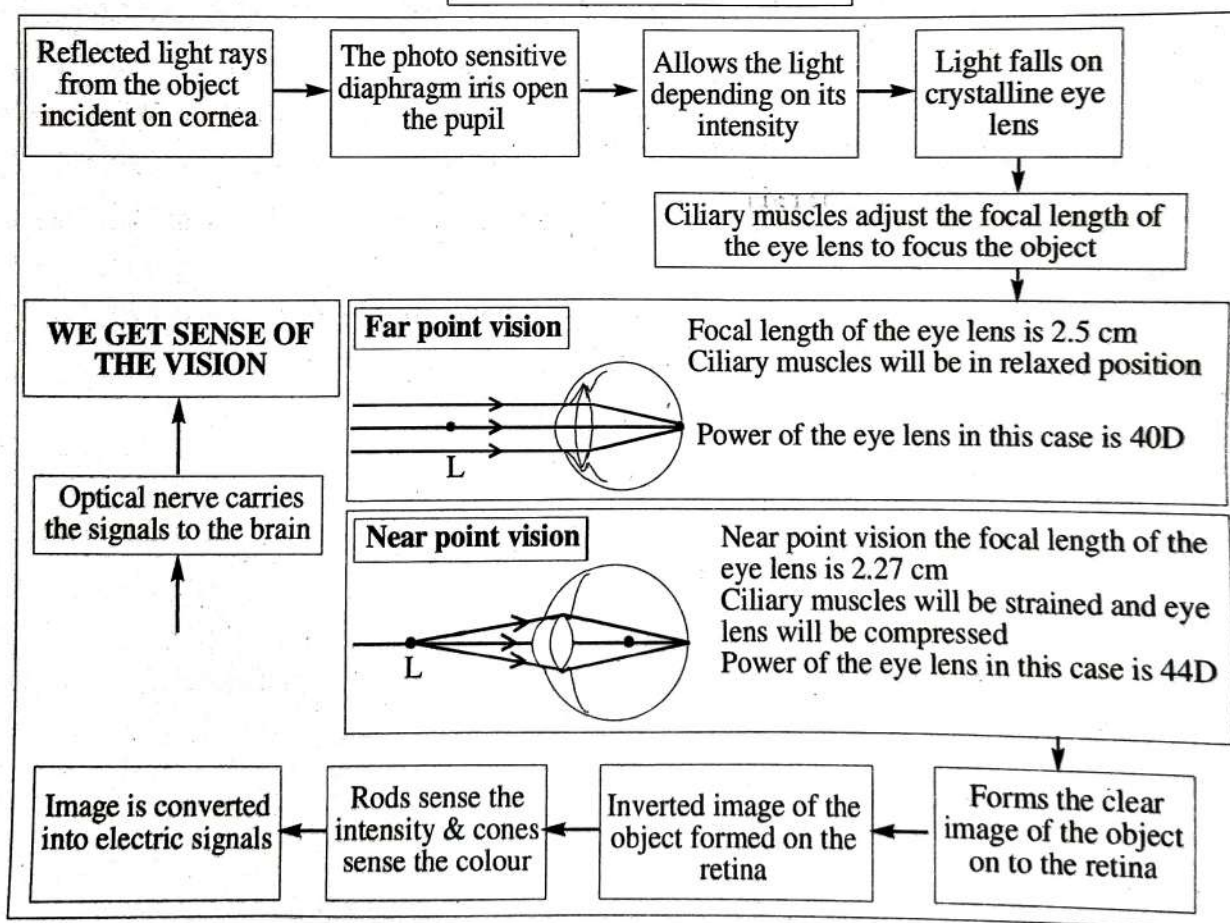
Rods : Rods help to detect the intensity of light

Cones : Cones help us to identify the colour

Optic nerves : These are the nerves which send the electrical signals generated by the illumination due to activation of light sensitive cells.

2.

WORKING OF HUMAN EYE



3.

DEFECTS OF VISION

Defects of vision occur in two types

When ciliary muscles loose their exapanding and contracting ability

Due to malnutrition, straining to read dul light, lack of vitamins

Myopia or near sightedness

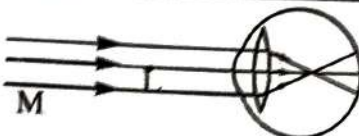


Normal eye : Image for far point object forms on the retina

Hypermetropia or Far sightedness



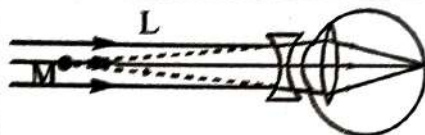
Normal eye : Image for near point object forms on the retina



Myopia (Near sightedness) :
A person can see only the nearer objects, unable to see the distant objects at far point
The maximum focal length of eye lens will be less than 2.5 cm
Power of the eye lens will be greater than 40D
Image forms infront of the retina



Hypermetropia (Far sightedness) :
A person can see only the distant objects clearly but unable to see the objects at near point
Minimum focal length is greater than 2.27 cm
Power of the eye lens will be less than 44D
Image formes beyond the retina



Correction : It can be corrected by using bi-concave lens



It can be corrected by using bi-convex lens

Presbyopia

This is happend due to ageing
Accommodation of the eyelens decreases
This can be corrected by using bi focal lens

4.

REFRACTION OF THE LIGHT THROUGH THE PRISM



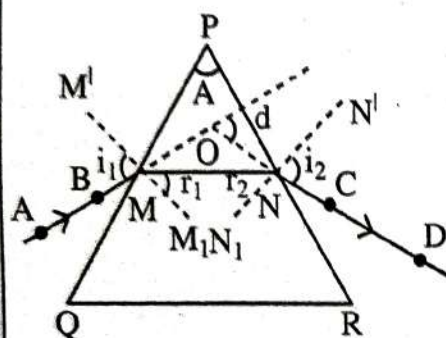
Prism



Prism

When light ray incident on the surface of the prism after refraction it will bend towards the base

In minimum deviation position incident light ray will travel parallel to the base after refraction in side the prism



Angle of the prism : The angle between two lateral surfaces

Angle of emergence (i_2) : The angle between the emergent ray and normal

Angle of the prism : The angle between the plane surfaces PQ and PR

Angle of deviation (d) : The angle between the extended incident ray and emergent ray

The relation between i_1, i_2 is $i_1 + i_2 = A + d$

The relation between r_1, r_2 is $r_1 + r_2 = A$

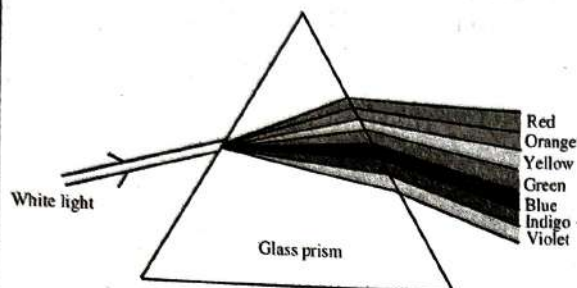
The deviation of the prism is $d = i_1 + i_2 - A$

The deviation will become minimum if $i_1 = i_2$ and $r_1 = r_2$

From Snell's law refractive index $n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

5.

Dispersion of white light through a glass prism



The ray theory could not explain the dispersion of white light because light has chosen seven paths of least time after refraction through the prism.

Wave theory explains the dispersion correctly.

The splitting of white light into seven colours after passing through the glass prism is called dispersion

The pattern of seven colours obtained is called spectrum

White light is a collection of waves with different wavelengths. Wave length of violet is less than wave length of red

During the separation of colours, producing a spectrum, refractive index decreases with an increase in wave length.

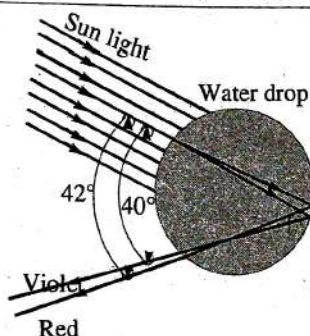
The refractive index of red is low hence it suffers low deviation.

Different colours of light rays undergo refraction at two surfaces then they will get different angles of deviation and get separated

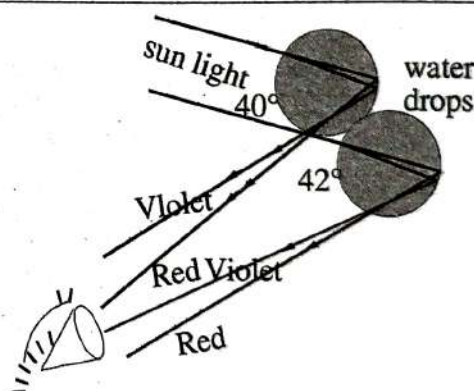
If we send a single colour ray through the prism then frequency doesn't change due to refraction. And remain unchanged

6.

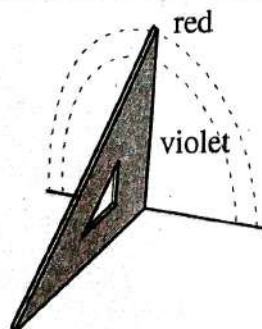
FORMATION OF RAINBOW



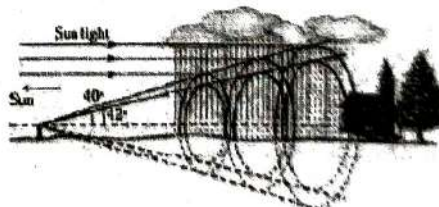
Rainbow : The spectacular pattern of colours of light formed due to dispersion of sun light passing through the tiny water drop lets present in the atmosphere after rain fall is called rainbow



Violet colour deviates more, Red colour deviates less
Angle of vision for Violet is 40° and for Red 42°



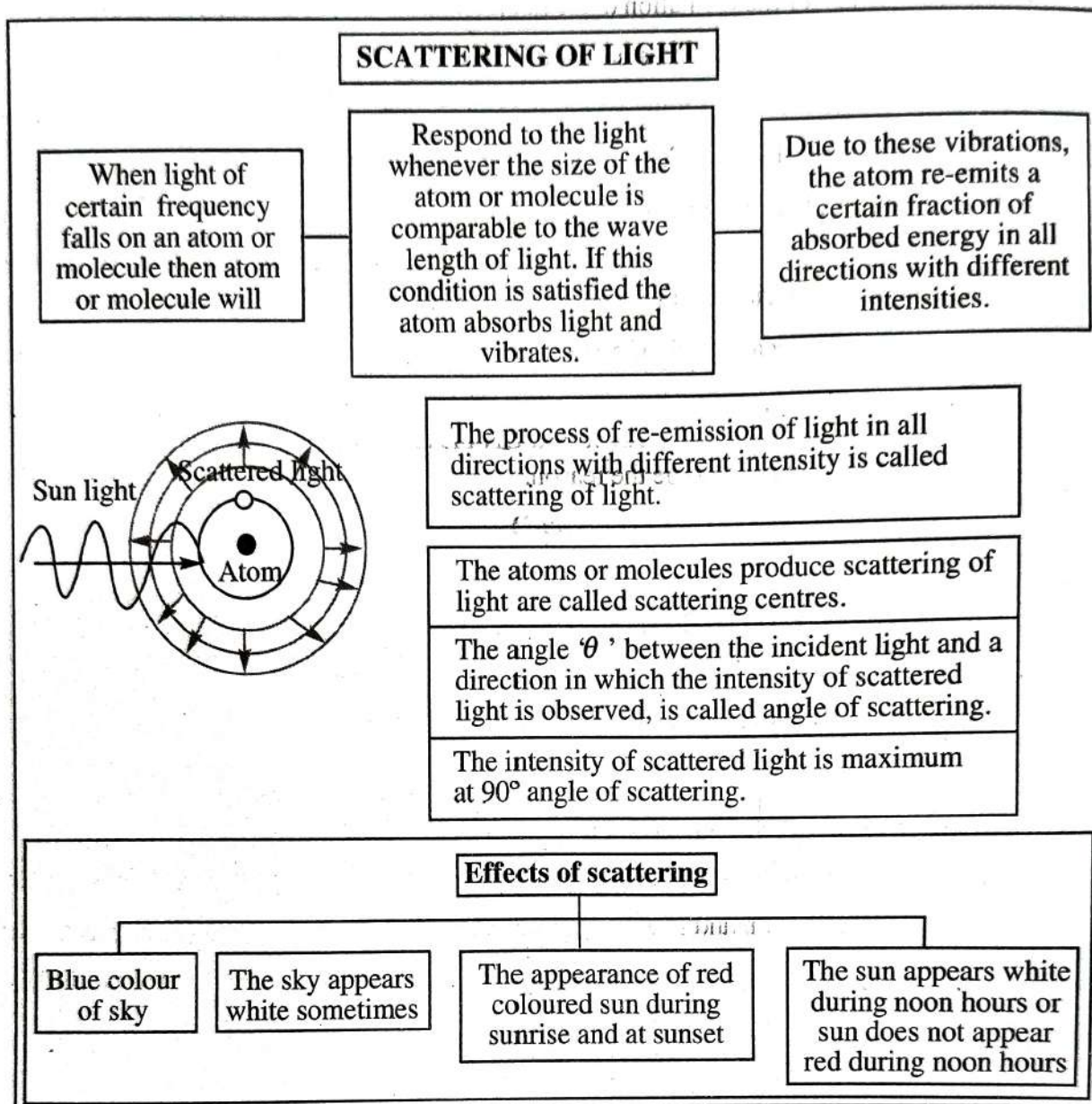
In between 40° and 42° we can observe the remaining colours of VIBGYOR



Rainbow is not the two dimensional arc, it is in the shape of three dimensional cone

During the formation of rainbow light ray will undergo two refractions and a total internal reflection in water drop

7.

**PRIORITY - I****Conceptual understanding (AS₁)**

1. What is meant by least distance of distinctive vision ? (Part- I. Pg.No. 184)
2. What are rods and cones? Give their action. (Part- I. Pg.No. 164)
3. What is meant by Myopia ? Write the specification. (S.S.C - 2018, 2019) (Part- I. Pg.No. 163)
4. What is meant by Hypermetropia ? Write the specification. (Part- I. Pg.No. 164)
5. Explain the correction of Hypermetropia. (T.Q) (Part- I.Pg.No. 147)
6. How do you correct the eye defect Myopia ? (T.Q) (Part- I.Pg.No. 147)
7. Explain the formation of rainbow. (S.S.C - 2019) (T.Q) (Part- I.Pg.No. 149)

8. Explain two activities for the formation of artificial rainbow. (T.Q) (Part- I.Pg.No. 150)
9. Explain briefly the reason for the blue colour of the sky ? (S.S.C - 2015) (T.Q) (Part- I.Pg.No. 166)
10. What is the scattering of light ? Explain with the help of an example. (Part- I. Pg.No. 162)
11. What is the cause of Presbyopia ? (S.S.C - 2015) (Part- I. Pg.No.185)
12. Draw the diagram of a lens which will be recommended by an eye doctor to a long - sighted patient. (C.C.E)(Part- I. Pg.No.171)
13. Define power of a lens .Write its unit (Part- I. Pg. No . 186)
14. How do you decide the focal length of convex lens required to correct the defect of hypermetropic eye? (S.S.C - 2017) (Part- I. Pg.No. 169)
15. Surya wakeup at night 12'o clock and switched ON the tube light. He is unable to open the eyes in that brightness. Guess what could be the reason. (C.C.E) (Part- I. Pg.No. 173)

(OR)

We will not able to see the things clearly when we come out of a darkroom, why?

16. Name the phenomenon associated with the following : (C.C.E)(Part- I.Pg.No. 175)
 - a) The sky appears blue
 - b) Formation of a rainbow in the sky
 - c) Twinkling of stars

Asking questions and making hypothesis (AS₂)

1. What is meant by dispersion ? When will it take place ? (C.C.E) (Part- I.Pg.No. 171)
2. What happens the dispersion and scattering of light do not occur ? Guess and write. (S.S.C - 2018). (Part- I.Pg.No. 163)

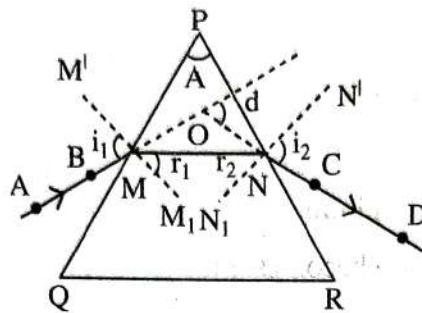
Experimental skills and field investigation (AS₃)

1. How do you find experimentally the refractive index of material of a prism. (T.Q) (Part- I.Pg.No. 147)
2. Suggest an experiment to produce rainbow in your class room and explain procedure. (T.Q) (Part- I.Pg.No. 153)

Communication through drawing, diagrams and graphs (AS₅)

1. Draw the ray diagram to show the path of the ray incident on a equilateral prism travelling with minimum deviation. (C.C.E) (Part- I. Pg.No. 171)
2. Draw the neat diagram depicting myopia and its rectification. S.S.C - 2024 (Part- I. Pg.No. 181)
3. Draw the neat diagram depicting hypermetropia and its rectification. (T.Q)(Part- I. Pg.No. 101)
4. Draw a ray diagram to show the angle of deviation when a ray of light passes through a glass prism. (S.S.C - 2015) (Part- I. Pg.No. 148)

5. Name and explain the given terms denote when light ray is passing through the prism ? (C.C.E)
 AB, MN, CD, $\angle A$, $\angle i_1$, $\angle r_1$, $\angle i_2$, $\angle d$ (Part- I. Pg.No. 171)



6. When a ray of white light incident on a prism then it will make seven colours to emerge out. draw the neat diagram depicting the given information. (C.C.E) (Part- I. Pg.No. 172)

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

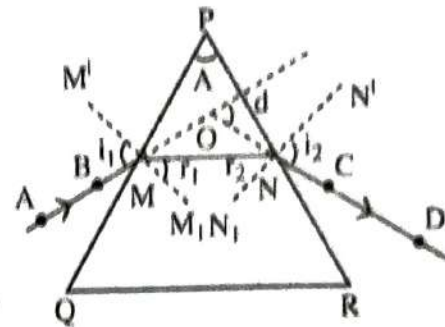
- How do you appreciate the working of ciliary muscles in the eye ? (S.S.C - 2018, 2022)
(T.Q) (Part- I. Pg.No. 166)
- How do you appreciate the role of molecules in the atmosphere for the blue colour of the sky ?
(T.Q) (Part- I. Pg.No. 166)
- Why light splits into different colours when it passes through a prism ? (Part- I. Pg.No. 187)
- Glass is known to be transparent material. But ground glass is opaque and white in colour. Why ?
(T.Q) (Part- I. Pg.No. 152)
- Why does sky appears sometimes white ? (T.Q) (Part- I. Pg.No. 151)
- Write two situations in day to day life where we observe dispersion of light ? (S.S.C - 2018) (C.C.E)
(T.Q) (Part- I. Pg.No. 172)

PRIORITY - II

Conceptual understanding (AS₁)

- What is the reason for sun does not appear red during noon hours ? (Part- I. Pg.No. 165)
- A prism with an angle $A = 60^\circ$ produces an angle of minimum deviation of 30° . Find the refractive index of material of the prism ? (Part- I. Pg.No. 146)
- Why does the sun appear reddish early in the morning? Explain with a diagram.
(Part- I. Pg.No. 166)
 - Why does the sky appears dark instead of blue for astronauts ?
- In which conditions does a rainbow form ? Why ? (C.C.E) (Part- I. Pg.No. 172)
- The focal length of a lens suggested to a person with Hypermetropia is 100 cm. Find the distance of near point and power of the lens. (T.Q) (Part- I. Pg.No. 153)

6. In the adjacent diagram the refraction of light ray which is incident on one of the inclined surface is depicted then find the relation among the followings (C.C.E) (Part- I. Pg.No. 172)



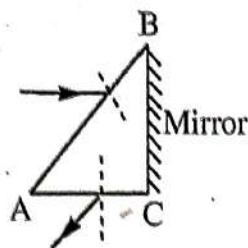
- The relation between i_1, i_2
 - The relation between r_1, r_2
 - The expression for deviation d
 - Draw the $i - d$ graph
 - When will deviation become minimum?
 - What happens when deviation becomes minimum?
 - Deduce relation for refractive index of material of the prism.
 - Draw the diagram describing the refraction of light ray through a glass prism with minimum deviation.
7. Distinguish between dispersion and scattering of light. (C.C.E)(Part- I.Pg.No. 174)
8. Doctor advised to use 2D lens. What is its focal length? (Part- I. Pg.No. 146)

Experimental skills and field investigation (AS₃)

- How can you say that human eye will need minimum distance for distinct vision? (C.C.E) (Part- I. Pg.No. 173)
- Describe an activity to find the angle of vision of human eye? (C.C.E) (Part- I. Pg.No. 174)

Communication through drawing, diagram and graphs (AS₅)

- Draw the ray diagram describing how sunlight disperses from rain drops reaches the eyes of observer? (Part - I. Pg.No. 150)
- Incident ray on one of the face (AB) of prism and emergent ray from the face AC are given in the figure. Complete the ray diagram. (T.Q) (Part - I Pg.No. 152)



- Describe the parts of human eye and its working with neat diagram? (Part - I. Pg.No. 167)

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

- Eye is the only organ to visualise the colourful world around us. This is possible due to accommodation of eye lens. Prepare six line stanza, enlighten your wonderful feelings. (T.Q) (Part- I.Pg.No. 153)

PRIORITY - III

Conceptual understanding (AS₁)

1. A boy uses spectacles of focal length – 60 cm. Name the defect of vision he is suffering from. Which lens is used for the correction of this defect ? Compute the power of this lens.
(C.C.E) (Part-I Pg.No. 174)
2. How can you say dispersion ruled out ray theory. (C.C.E)(Part- I Pg.No. 174)
3. How do you find the maximum and minimum focal lengths of eye lens ? (Part- I Pg.No. 168)
4. How do you decide the focal length of concave lens required to correct the defect of myopic eye ?
(Part- I Pg.No. 169)
5. Why does it take some time to see the objects in a dim room when we enter the room from bright sunlight outside?
(Part- I Pg.No. 161)
6. What is meant by scattering center ? (Part- I Pg.No. 162)
7. When will the intensity of scattered light is maximum ? (Part- I Pg.No. 162)
8. Why does the light dispersed by the rain drops appear as a bow ? (Part- I Pg.No. 165)
9. Doctor identified the maximum focal length of Ramesh's eye is 2.499 cm. (Part- I Pg.No. 175)
 - a) What type of visual problem he has?. (C.C.E)
 - b) What type of lens is needed for his problem ?
 - c) Draw a neat diagram after rectifying of eye vision.
10. What is the reason behind to form VIBGYOR when white light is incident on a prism.
(C.C.E) (Part- I Pg.No. 175)

Asking questions and making hypothesis (AS₂)

1. A prism can disperse the white light into seven component colours, where as a glass slab cannot, why? Justify your answer. (C.C.E) (Part- I Pg.No. 175)
2. Guess and write what happen if power of lens (Diopetre) is not come into useful.

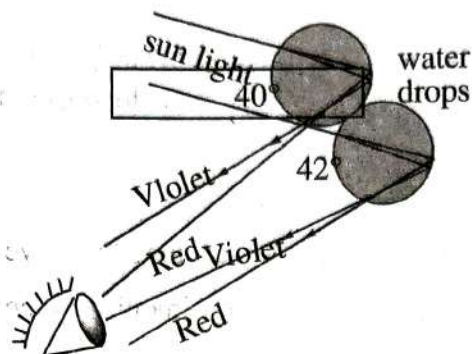
Experimental skills and field investigation (AS₃)

1. How do you demonstrate scattering of light by an experiment ? (Part- I Pg.No. 154)

Information Skills (AS₄)

1. A glass slab is placed over a page on which the word VIBGYOR is printed with each letter in its corresponding colour (Part- I Pg.No. 166)
 - i) Will the image of all the letters be in the same place ?
 - ii) If not, state which letter will be raised to the maximum. Give a reason for your answer

2. Observe the following diagram.

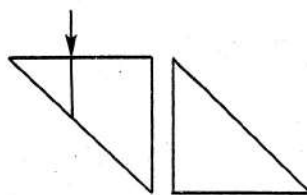


Answer the following questions :

- What information does the above picture give ?
- What will be the angle between an incident light ray and emergent light ray from the water drop to see red colour with our naked eye?
- What will be the angle between incident and emergent rays to see all colours together ?
- What happens when a light ray from the sun falls on water droplets ?

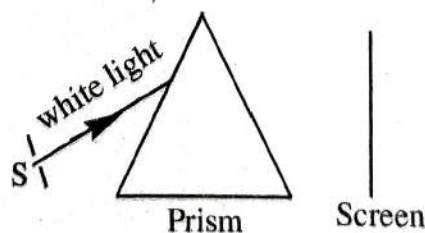
Communication through drawing, diagrams and graphs (AS₃)

- Two isosceles right - angled glass prisms are placed near each other as shown in the figure. (The critical angle for glass is 42°). Complete the path of the light ray entering the first isosceles right - angled glass prism till it emerges from the second identical prism. (C.C.E) (Part- I, Pg.No. 176)



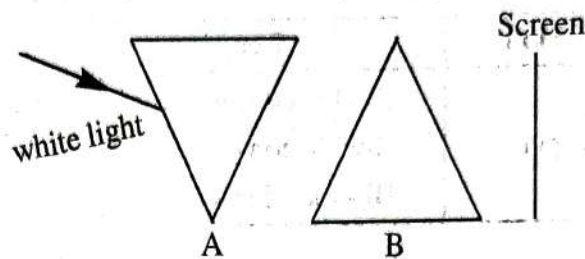
- Figure shows a thin beam of white light from a source S striking on one face of prism.

(C.C.E) (Part- I, Pg.No. 176)

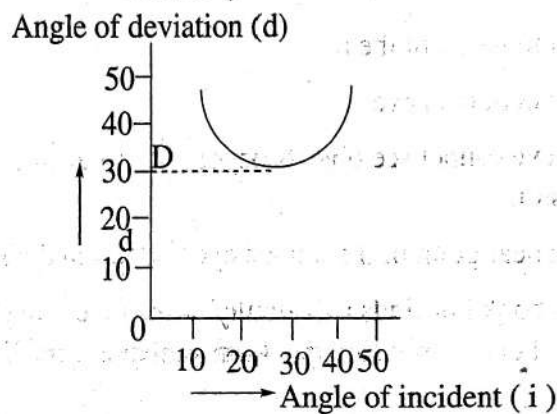


Complete the diagram to show the effect of prism on the beam and to show what is seen on the screen.

3. Complete the ray diagram given below to show the nature of light produced on the same. (C.C.E) (Part- I. Pg.No. 177)



4. Raju conducted an experiment with a prism having 60° angle of prism and drawn a graph as shown below. Find the refractive index of the prism? (C.C.E)(Part- I. Pg.No. 177)



Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

1. A rainbow viewed from an airplane may form a complete circle. Where will the shadow of the airplane appear? Explain. (Part- I. Pg.No. 165)
2. Phani's grand father is unable to read the newspaper, watching this phani gave a lens to his grand father and asked to try reading the news paper using the lens. (Part- I. Pg.No. 178)
 - a) Here what kind of lens phani gave? (C.C.E)
 - b) Write the reasons for doing so. Try to explain with the help of diagram.

SESSIONWISE STUDENT TASK

1. What part of the eye controls light that enters in to eye. How does it work ?
2. The following table is based on defects of vision in eye ball.

Sl.No.	Defect of vision	Lenses
1.	Myopia	Double concave lens
2.	Hyper metropia	Double convex lens
3.	Prebyopia	Bi - focal lenses

- i) Write the name of defect of vision a person cannot see object at long distances but can see nearby objects clearly and write the name of lens to rectify defect of vision.
 - ii) Write the name of defect of vision a person find it difficult to see nearby objects clearly and distinctly and write name of the lens to rectify defect of vision.
3. Explain power of accommodation of the eye?
 4. Write one function of iris in human eye.
 5. A person with a myopic eye cannot see objects beyond 1.2 m distinctly. Which type of lens is to be used to restore proper vision?
 6. Write about far point and near point of the human eye with normal vision?
 7. I can comfortably read a book but finds a difficulty to read the number on a bus parked 5 m away from me. Name the type of defect of vision that I am suffering from. Which type of lens should I use to correct my vision?
 8. If we increase the distance of an object from the eye, what happens to the image distance in the eye?
 9. What happens if ciliary muscles does not function well?
 10. A 14-year old student is not able to see clearly the questions written on the blackboard placed at a distance of 5 m from him.
 - a) Name the defect of vision he is suffering from.
 - b) Name the type of lens used to correct this defect.
 11. Which coloured light will travel the slowest in glass ?
 12. Write two observations in the experiment to find the refractive index of prism.
 13. The minimum focal length of an eye lens is more than 2.27 cm. Mention the eye defect and its correction.
 14. Chandraiah consulted an "eye specialist" (ophthalmologist) to rectify his deffect of vision. Then doctor suggested chandraiah to use a concave lens. Draw the ray diagram to expain the chandraiah's defect of vision and also the diagrams explaining how concave lens help to correct his defect of vision?

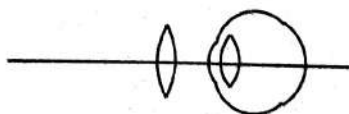
15. Ravi and Ramu suffered from the eye problem and went to the doctor. The doctor suggested that they use the following types of power of lens. Observe the information and answer the following questions.

Persons	Power of lens suggested by Doctor (in Diaptr)
Ravi	+ 1.5 D
Ramu	- 1.5 D

- What is the defect of vision of Ravi ?
 - Who cannot clearly see the objects that are away ?
 - What lens was used to prevent the defect of vision of Ramu ?
 - What is the focal length (in metre) of the lens used to prevent Ravi's defect of vision ?
16. Complete the path of white light incident on water drop



- Write the importance of "Graph" in finding the refractive index of prism.
- You are travelling in a bus beside a lake. Water is sprinkling from a fountain. You observe a rainbow. But after travelling some distance it was not seen. How do you explain it ?
- Why we should close our eyes, when a student focussed a bright light to our eyes ?
- Write any two precautions that are taken while finding the refractive index of the prism ?
- Identify the below diagram. Complete the diagram and write the nature of it.



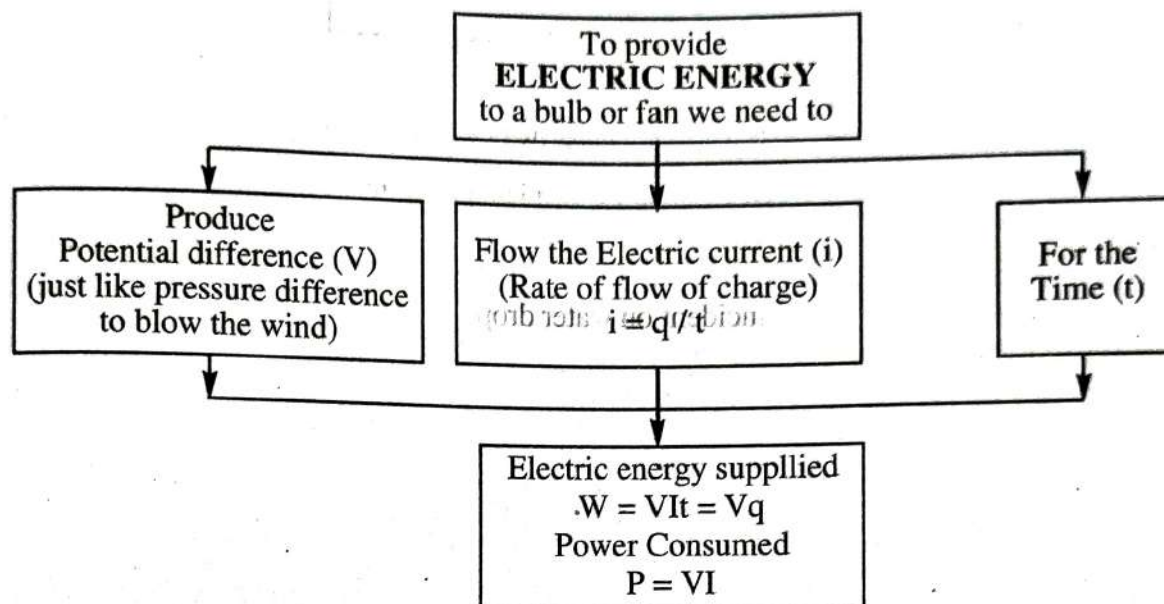
- Do you know the reasons for appearance of red coloured sun during sunrise and sunset ?

(S.S.C - 2023)

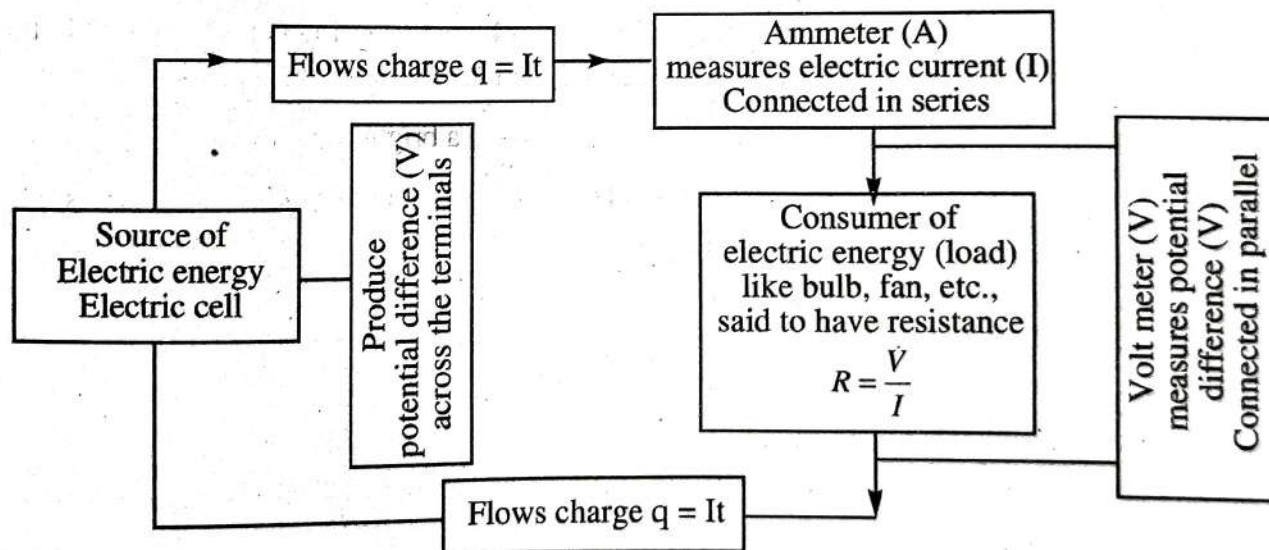
4. ELECTRIC CURRENT

SUMMARY

1.



2.

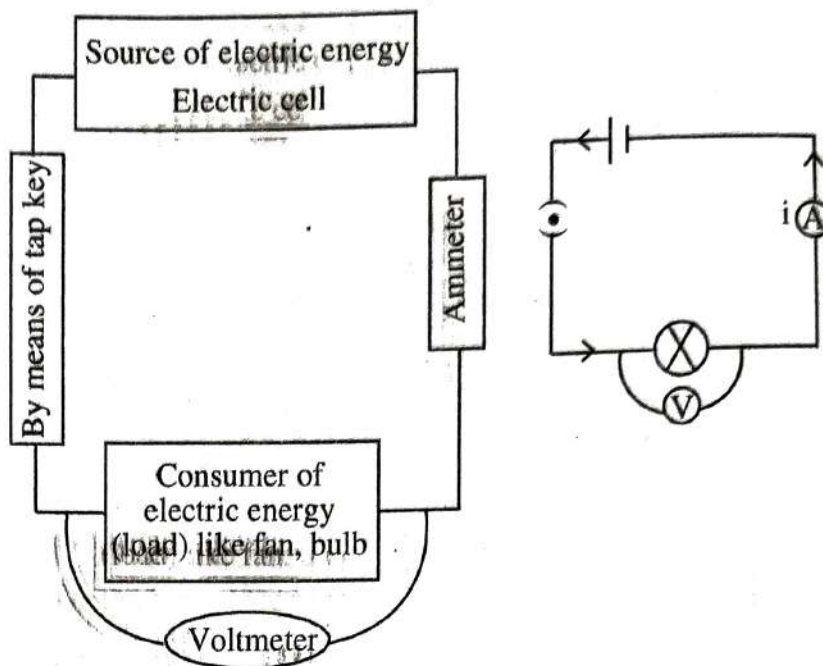


3.

ELECTRIC CIRCUIT
closed path connecting
the appliances with source
of electric energy

Connecting in the same
path along the circuit is
referred as series
connection or **series
combination**

Connecting across same
terminals or between
same two points along the
circuit is referred as
**parallel connection or
parallel combination**

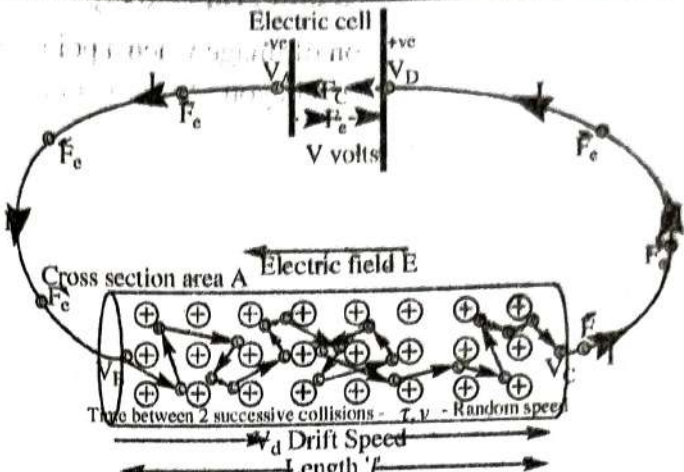


4.

WORKING OF ELECTRIC CELL

CHARGING Open circuit Electrolyte Electrolyte $F_c = F_e$ in put $EMF \epsilon = \frac{W_e}{q}$	→ Converts chemical energy into electric energy → Electrolyte produce chemical force F_c → F_c Separate the charges on to the opposite plates → F_e is produced between separated charges → Charging takes place until $F_c = F_e$ → We rate the work done for separating unit charge by F_c as $e.m.f. \quad \epsilon = \frac{W_e}{q}$ → e.m.f. = potential difference in open circuit
DISCHARGING Electrolyte Electrolyte $F_c > F_e$ out put Potential difference $V = \frac{W_e}{q}$	Closed circuit → Conductor is connected across the terminals → F_e drives the charge carriers from one plate to another along the conductor → Charge on two plates decreases so $F_e < F_c$ → F_c starts charging again until $F_c = F_e$ → As we move from one terminal to another terminal the potential of the charge decreases from V to zero

5.



Conductor is connected to the electric cell of e.m.f. $\mathcal{E}$

Potential difference produced by the cell causes F_e

F_e accelerates the charge carriers

Accelerated charge carriers pickup random speed

Moving charge carriers collide with lattice ions and come to rest

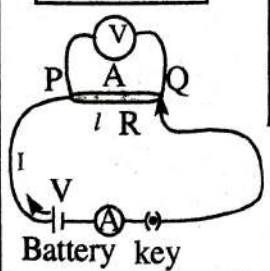
Again F_e accelerates charge carriers, they come to stop after collision. This process repeats again and again charge carriers are conducted with drift speed v_d

Drifting carriers establish electric current $I = neAv_d$

CONDUCTION OF CHARGE
through the conductor containing free electron density (n)

6.

The current through the circuit depends on



Battery key

Potential difference applied V , $V = I \times R$ (ohm's law)
 R = electric resistance of conductor

Length of the conductor l

Time interval between two successive collisions τ

Cross section area of conductor A

Free electron density n

Electric resistance

$$R = \frac{V}{I}$$

Electric shock is a combined effect of potential difference, electric current and resistance of the human body.

- When human body applied with potential difference 240 volt mains.
- Small current starts flowing through the human body due to its high resistance initially.
- As we continue to flow the current then resistance of the body starts decreasing.
- Gradually current value increases and it causes painful affects and becomes fatal referred as electric shock.

7.

Every conductor or load offer resistance.

Resistance of a conductor depends on

Nature of material

Different materials contain different n , values τ

Temperature of the conductor as temperature changes τ value changes due to additional thermal energy

Length of the conductor $R \propto l$

Cross section area of the conductor $R \propto \frac{1}{A}$

Resistance of given material having unit length and unit cross section area is called resistivity or specific resistance $\rho = \frac{RA}{l}$

8.

CONDUCTORS

Metals are examples of conductors due to their low resistivity

Alloys are used to make heating elements in toaster, electric iron

They do not oxidise easily

Their resistance vary very little with temperature

Tungsten is used as filament in electric bulb due to high resistivity and melting point

Fuse wire is used to prevent over loading due to its low melting point

Basing on resistivity materials are classified

$$\rho = \frac{RA}{l} \Omega\text{-m}$$

INSULATORS

Ex : Glass, porcelene, wood, plastic etc.,

They possess larger values of resistivity than metals

The resistivity of insulators is very high of order 10^{14} to $10^{16} \Omega\text{m}$

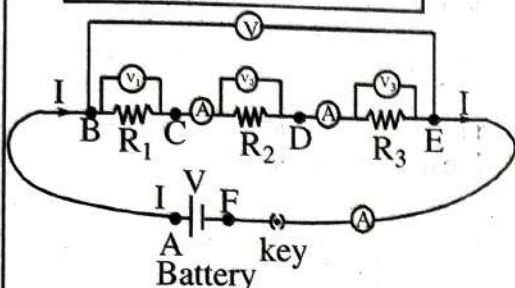
SEMI CONDUCTORS

The resistivity of semiconductors are 10^{15} to 10^{16} times less than that of insulators

Silicon and germanium are examples of semi conductors

9. Solving Electric Circuits :

Analyzing the single loop circuit parameters. Along closed loop we have to



1. $V_A = V_B > V_C > V_D > V_E = V_F$, $V_A = V$, $V_F = 0$
2. $V_{BC} = V_1$, $V_{CD} = V_2$, $V_{DE} = V_3$, $V_{BE} = V$
 $V = V_1 + V_2 + V_3$
3. $V_{BC} = IR_1$, $V_{CD} = IR_2$, $V_{DE} = IR_3$

Track the potential of the electric charge at every point

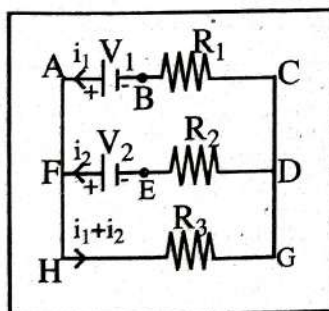
Monitor the current(I) value at every point

Measure the potential drop between two successive points as $V = IR$

Verify ohm's law across two successive points

Analyse resistance across any two points by taking the ratio between potential drop (V) across those two points and current (I) $R = \frac{V}{I}$

Analyzing the multi loop circuit parameters. Kirchhoff's laws



Along closed loop the total potential difference becomes zero. (Law of conservation of energy)

At every junction we observe algebraic sum of currents reaching the junction is zero. (Law of conservation of charge)

1. From the ABCDEFA loop we get
 $-V_1 + V_2 + i_1 R_1 - i_2 R_2 = 0$

2. From the AFHGDCBA loop we get
 $-R_3 \times (i_1 + i_2) - i_1 R_1 + V_1 = 0$

3. From the FEDGH loop we get
 $-V_2 + i_2 R_2 + R_3 \times (i_1 + i_2) = 0$

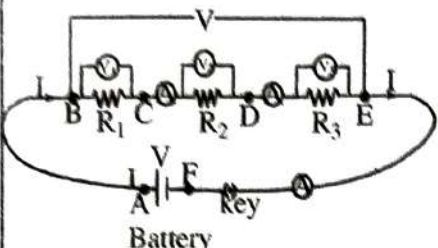
4. At junction F $i_1 + i_2 = i$ through HG

5. At junction D $i_1 = i$ through HG $- i_2$

6. On solving the above equations we get all parameters

10. Series combination of resistances or loads :

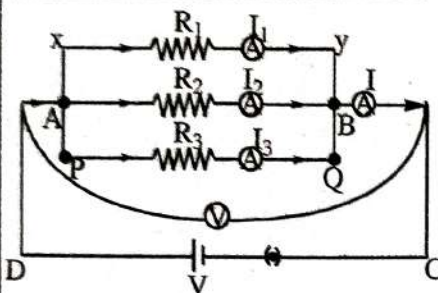
In series combination of resistances or loads



- Connecting the loads in the same path along the circuit is referred as series combination.
- Same current(I) will flow through all resistances
- Total voltage will be divided across each and every individual resistance
 $V = V_1 + V_2 + V_3$
- The total resistance is greater than any of the individual resistance
 $R = R_1 + R_2 + R_3, R > (R_1, R_2, R_3)$
- The total P.d across the closed loop ABCDEFA upto the starting point is zero
 $V_A - V_{BC} - V_{CD} - V_{DE} + V_{FA} = V_A$
 $V_A - V_{BC} - V_{CD} - V_{DE} = V_F = 0$

11. Parallel combination of three resistors :

In parallel combination of three resistors R_1, R_2, R_3 ($R_1 < R_2 < R_3$) we observe



- Connecting the loads across same two terminals or points along the circuit is referred as parallel combination.
- Same p.d (V) is applied across each and every resistors
- $I_1 : I_2 : I_3 = R_3 : R_2 : R_1$ Total current $I = I_1 + I_2 + I_3$
- $I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$ According to Ohm's law
- The total resistance of the combination is less than any of the individual resistances
- The total resistance R of the combination is such that

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

12.

Utilization of electric energy

- Charge flows from higher potential to lower potential
- Upon moving from -ve terminal to +ve terminal along the conductor the potential of the charge becomes zero
- The total potential drop across the closed loop of a circuit is zero
- Flowing charge releases energy
- Energy dissipated across the load $W = Vit = i^2 Rt = \frac{V^2 t}{R}$ (Joule / kWh)
- Power consumed by the load $P = VI = \frac{V^2}{R} = I^2 R$ (watt)
- When excess of energy flows through the circuit it leads to over heating of appliances this is called overloading
- when source terminals get direct contact then also excess energy flows this is referred as short circuit
- to prevent this over loading and short circuit we use a metal wire with low melting point across the circuit called fuse wire.

PRIORITY - I

Conceptual understanding (AS₁)

1. What is the value of 1 kWh in joules ? ((Part-II. Pg.No. 42)
2. Write the differences between potential difference and emf. (T.Q) (Part-II. Pg.No. 34)
3. Explain how electron flow causes electric current with Lorentz - Drude theory of electrons. (T.Q) (Part-II. Pg.No.33)
4. What are the factors affecting the resistance of an electric conductor ? Explain any two factors.
(or) Derive $R = \frac{\rho l}{A}$ (S.S.C - 2018) (T.Q)(Part- II Pg.No. 74,42)
5. Deduce the expression for the equivalent resistance of three resistors connected in series. (Part-II. Pg.No. 47)
6. Silver is better conductor of electricity than copper why do we use copper wire for conduction of electricity. (T.Q) (Part-II. Pg.No. 42)
7. Deduce the expression for the equivalent resistance of the three resistors connected in parallel ? (Part-II. Pg.No. 48)
8. What do you mean by electric shock ? Explain how it takes place. (T.Q) (Part-II. Pg.No. 34)
9. Explain overloading of household circuit. (T.Q) (Part-II. Pg.No. 35)
10. Why do we use fuses in household circuits ? (S.S.C - 2017) (T.Q) (Part-II. Pg.No. 35)
11. Three resistors $R_1 \Omega$, 4Ω , and 8Ω , are connected in series in a circuit. If the resultant resistance in the circuit is 20Ω find the value of R_1 .
12. State ohm's law? (Part-II. Pg.No. 37)
13. Write the limitations of ohm's law ? (Part-II. Pg.No. 43)
14. What is electric current ? Write formula and S.I. unit for it ? (Part-II. Pg.No. 41)
15. What is meant by electric power ? Write its expressions ? CCE (Part-II. Pg.No. 75)
16. Why doesn't crow get a shock when it stands on a high Voltage wire ? (Part-II. Pg.No. 54)
17. Write the steps of sign convention to be followed while applying Kirchhoff's laws to the multi loop circuits. (Part-II. Pg.No. 45)
18. Three resistors 4Ω , 12Ω , 3Ω are connected in series with 12 V battery. Find the current in the circuit (S.S.C - 2018) (C.C.E) (Part-II. Pg.No. 54)

Asking questions and making hypothesis (AS₂)

1. Are the head lights of a car connected in series or parallel ? Why ? (T.Q) (Part-II. Pg.No.36)
2. Why should we connect the electric appliances in parallel to house hold circuit ? What happens if they are connected in series ? (S.S.C - 2019) (T.Q) (Part-II. Pg.No. 36)
3. Why do we consider tungsten as a suitable material for making the filament of a bulb ? (T.Q) (Part-II. Pg.No. 36)

Experimental skills and field investigation (AS₃)

1. How can you verify that the resistance of a conductor is temperature dependent? (T.Q) (Part-II. Pg.No. 51)

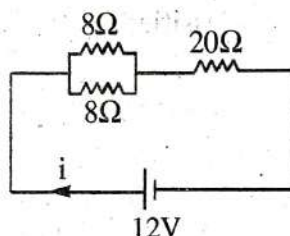
- How do you verify that resistance of a conductor is proportional to the length of the conductor for constant cross section area and temperature ? (S.S.C - 2015) (T.Q) (Part-II. Pg.No. 38)
- State Ohm's law. Suggest an experiment to verify it and explain the procedure ? (S.S.C- 2016, 2022, 2024) (T.Q) (Part-II. Pg.No. 37)

(OR)

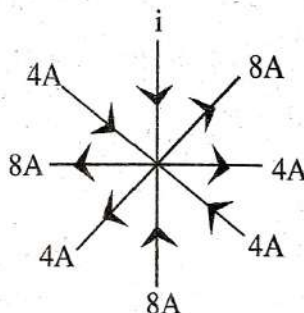
Describe an activity with the help a diagram to establish the relation between current (i) flowing in a conductor and potential difference (V) maintained across its ends. (C.C.E)

Communication through drawing diagrams of graphs (AS₃)

- Draw a circuit diagram for a circuit in which two resistors A and B are connected in series with a battery and a voltmeter is connected to measure the potential difference across the resistor A. (T.Q) (Part-II. Pg.No. 35)
- Find the resultant resistance and current flowing in the given circuit? (C.C.E) (Part-II. Pg.No. 54)



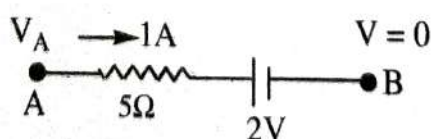
- Observe the given Junction and then find the value of i . (C.C.E) (Part-II. Pg.No. 55)



- Draw a circuit with battery, voltmeter, ammeter, resistor, connected with wires. (S.S.C - 2018) (Part-II. Pg.No. 55)

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

- How can you appreciate the role of a small fuse in house wiring circuit in preventing damage to various electrical appliances connected in the circuit ? (T.Q) (Part-II. Pg.No. 46)
- A uniform wire of resistance 100Ω is melted and recast into wire of length double that of the original. What would be the resistance of the new wire formed ? (Part-II. Pg.No.48)
- In the figure the potential at A is _____ when the potential at B is zero. (T.Q) (Part-II. Pg.No. 35)



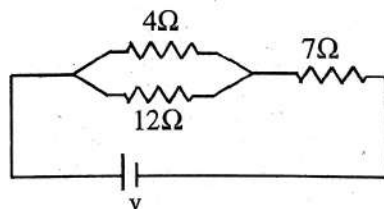
- Write two applications of Non ohmic conductor in day to day life situations.

(C.C.E)(Part-II. Pg.No. 55)

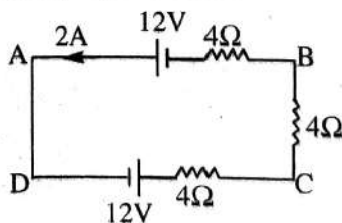
PRIORITY - II

Conceptual understanding (AS₁)

- How does a battery work? Explain. (T.Q) (Part-II. Pg.No. 46)
- Explain Kirchhoff's laws with examples. (T.Q) (Part-II. Pg.No. 47)
- A wire of length 1m and radius 0.1m has a resistance of 100Ω . Find resistivity of the material. (Part-II. Pg.No. 46)
- Two bulbs have ratings 100 W, 220 V and 60 W, 220 V which one has the greatest resistance? (T.Q) (Part-II. Pg.No. 35)
- What is resistance? Is the value of resistance the same for all materials explain? (C.C.E) (Part-II. Pg.No. 55)
- Define the term 'resistivity' of a material. (Part-II. Pg.No. 41)
- Distinguish between resistance and resistivity of a material.
- Basing on resistivity how materials are classified, discuss? (C.C.E) (Part-II. Pg.No. 56)
- Two resistors $R_1\Omega$, and 12Ω , are connected in parallel in a circuit. If the resultant resistance in the circuit is 3Ω find the value of R_1 .
- During the process of explaining about the construction and working of a battery discuss
 - What happens while charging of a battery? (C.C.E) (Part-II. Pg.No. 56)
 - What happens while discharging of a battery?
 - What do you observe about voltage across battery in an open circuit and in a closed circuit?
- Observe the given circuit and calculate the resultant resistance.



- A house has 3 tube lights, two fans and a Television. Each tube light draws 40W. The fan draws 80W and the Television draws 60W. On the average all the tube lights are kept on for five hours, two fans for 12 hours and the television for five hours every day. Find the cost of electric energy used in 30 days at the rate of Rs. 3.00 per kWh. (T.Q) (Part-II. Pg.No. 37)
- Write the loop equations 1) ABCDA, 2) ADCBA (C.C.E) (Part-II. Pg.No. 58)



Asking questions and making hypothesis (AS₂)

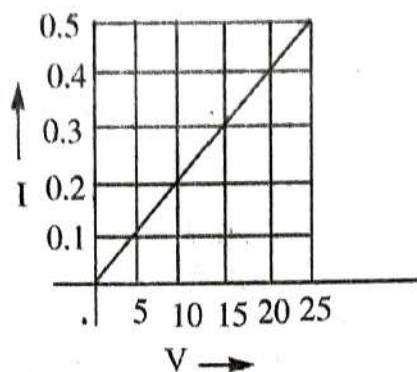
- Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Why does the nichrome wire get heated first? (C.C.E) (Part-II. Pg.No. 56)
- What happens if we use a fuse made up of same wire which is used to make the electric circuit? (2017 SSC)

Experimental skills and field investigation (AS₃)

1. How do you verify that resistance of a conductor is inversely proportional to the cross sectional area of the conductor at constant temperature and length of the conductor? (Part-II. Pg.No. 50)

Information skills (AS₄)

1. Sudhakar recorded several readings of voltage and currents across a material wire using voltmeter and ammeter in a table and the graph plotted using the values in the table is given as follows
(C.C.E) (Part-II. Pg.No. 57)



He represented the voltage (V) in volts and current (I) in amperes while plotting graph. Then answer the following basing on the information given in graph

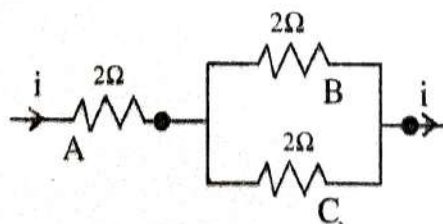
- What kind of material sudhakar used in this experiment.
- Calculate the resistance of material wire.
- Calculate the power consumed by the wire when a p.d of 20 V applied across it
- Write the principle present in the given graph

Communication through drawing diagrams of graphs (AS₅)

1. You have three resistors of values 2Ω , 3Ω and 5Ω . How will you join them so that the total resistance is more than 7Ω .
(C.C.E)(Part-II. Pg.No. 57)
- Draw a diagram for the arrangement.
 - Calculate the equivalent resistance.

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

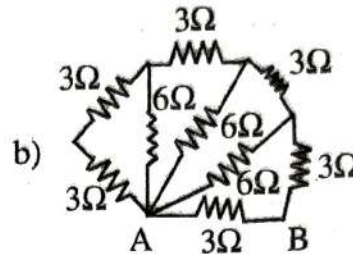
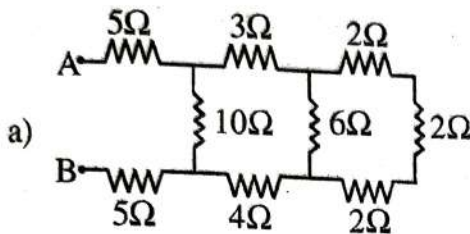
1. What precautions should be taken to avoid the overloading of domestic electric circuits?
(C.C.E) (Part-II. Pg.No. 43)
2. Three resistors A, B and C are connected as shown in the figure. Each of them dissipates energy to a maximum of 18 W. Find the maximum current that can flow through the three resistors.
(S.S.C - 2015) (Part-II. Pg.No. 58) (C.C.E)



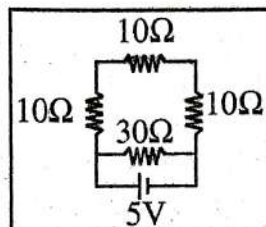
PRIORITY - III

Conceptual understanding (AS₁)

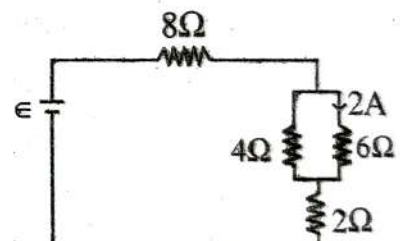
1. Find the equivalent resistance between A and B in the following case. (C.C.E) (Part-II. Pg.No. 58)



2. Calculate effective resistance of total circuit, Current and potential difference across the 30Ω Resistance. (C.C.E) (Part-II. Pg.No. 60)



3. Define drift speed and write the expression for it. (Part-II. Pg.No.61)
4. Observe the circuit and answer the following questions. (C.C.E)
- Calculate the current flowing through 4Ω resistor ?
 - What is the current drawn from the battery ?
 - What is the emf of the battery ? (Part-II. Pg.No. 71)



Asking questions and making hypothesis (AS₂)

1. Suppose that you have three resistors each of value 30Ω . How many resistors can you obtain by various combinations of these three resistors ? Draw diagrams in support of your predictions.

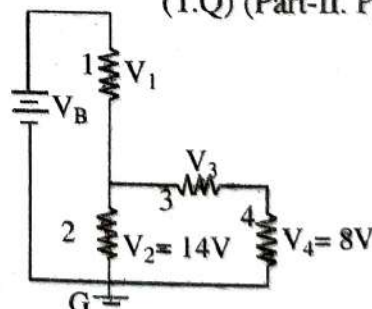
(T.Q) (Part-II. Pg.No. 36)

Aesthetic sense/ appreciation/ values (AS₃) / Application to daily life, concern to bio diversity (AS₄)

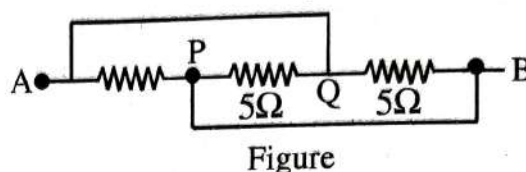
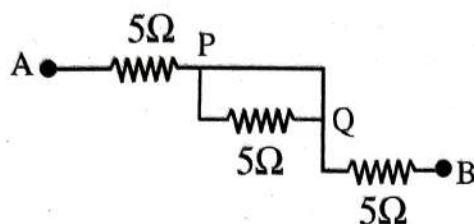
1. Observe the circuit and answer the questions Given below.

(T.Q) (Part-II. Pg.No. 53)

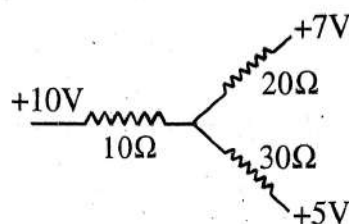
- Are resistors 3 and 4 in series ?
- Are resistors 1 and 2 in series ?
- Is the battery in series with any element ?
- What is potential drop across the resistor 3 ?
- What is the total emf in the circuit if the potential drop across resistor 1 is 6 V ?



2. Find the equivalent resistance between the points A and B. (S.S.C - 2017) (Part-II. Pg.No. 43)

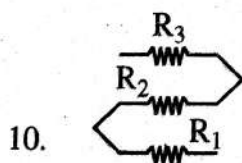


3. The current through 10Ω resistor in the figure is approximately (Part - II. Pg.No. 52)

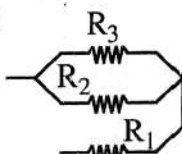


SESSIONWISE STUDENT TASK

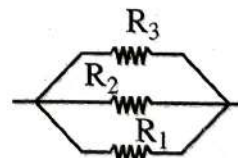
- What would happen if house hold electric components are connected in series ?
- What happens if filament in Electric bulb is made up of metals like aluminium, silver other than Tungsten ?
- What happens if a high melting point conductor is used as a fuse in house-hold connections?
- Do electric bulbs in houses and decorative bulbs are connected in the same way?
- In a house-hold circuit 220 voltage and 440Ω of resistance are connected, then find the electric current?
- Two resistances 4Ω & 8Ω are connected to get least resistance then find resultant resistance in the circuit?
- "Electric current increases as the emf of values increases in the circuit". Mention the apparatus and write experimental procedure ?
- What measures you will suggest to prevent the short circuit
- What happens, when the fuse in our home is connected in parallel to the electrical circuit ?



Circuit - 1



Circuit - 2



Circuit - 3

Among the above which circuit's resultant resistance is 2.5Ω when $R_1 = 1\Omega$, $R_2 = 2\Omega$ and $R_3 = 6\Omega$?

11. Observe the following information about resistivity of some materials and answer the questions.

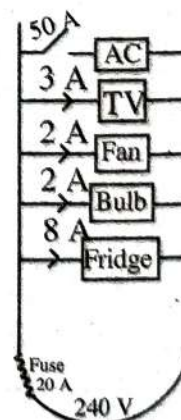
Material	Copper	Aluminium	Tungsten	Nichrome	Graphite	Silicon
$\rho(\Omega - m)$ at 20°C	1.68×10^{-8}	2.82×10^{-8}	5.60×10^{-8}	1.10×10^{-6}	2.50×10^{-6}	6.40×10^2

- Which of the above material is used to make domestic electric wiring ? Why ?
 - Which of the above can be used to make diodes and I.C's ? What are they called ?
12. Draw the neat schematic diagram representing household wiring that is how all electric appliances in your house are connected across the mains and also symbols used in the circuit connection?
13. Your friend required 10 ohm resistance. He came to you and asked for it. You have only resistors having 40 ohm then.
- At least how many resistors your friend seek?
 - What is the combination in which resistors are to be connected?
 - Show that the effective resistance of combination is 10Ω .
14. Resistivity of various materials are given in below table.

Material	Resistivity at 20°C
Gold	2.44×10^8
Tungsten	5.60×10^{-8}
Graphite	2.50×10^{-6}
Silicon	6.40×10^2
Rubber	1.00×10^{13}

Answer the following questions from the above table.

- Write the unit of resistivity.
 - What is resistivity value of filament in an electric bulb ?
 - Write the name of material used for making of semi conductors ?
 - Write how the resistivity changes from conductor to insulator.
22. Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Why does the nichrome wire get heated first?
23. Will current flow more easily through a thick wire or a thin wire of the same material when connected to the same source? Why?
24. An experiment reveals that "The ratio of V to I i.e., V/I is constant for a conductor." Answer the following questions relating to this point.
- Name the apparatus required for the experiment.
 - Tabulate the results in the experiment.
 - Write observations of the experiment.
 - If you draw graph of V & I , explain how it will be.
25. Observe the current consumption chart, and answer the following questions.
- What is the maximum load in the chart ?
 - What happens if AC button is ON ?



5. ELECTROMAGNETISM

SUMMARY

1.

OERSTED'S EXPERIMENT

Conductor carrying current
Electromagnet

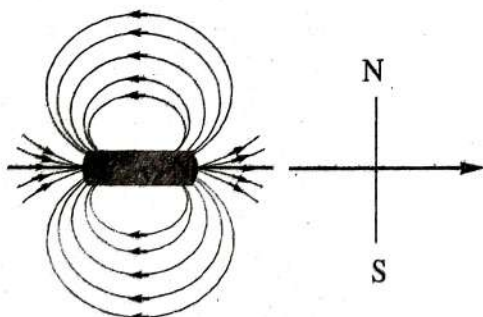
- Produce Magnetic field around it
- It influences compass needle to change its direction from North - South
- This phenomenon is called electromagnetism
- The strength of the field depends on strength of current
- The direction of field can be determined by using right hand rule

2.

Magnetic field

The region around the magnet upto which we can feel its influence
It is produced by lines of force called magnetic lines of force

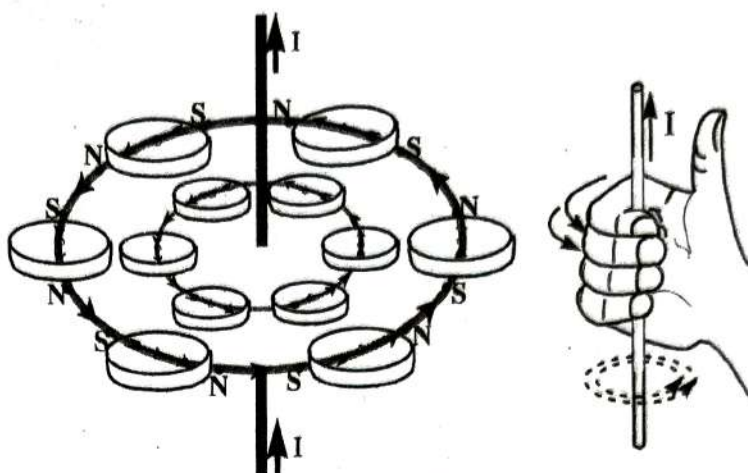
Magnetic flux density $B = \frac{\phi}{A}$
S.I unit of magnetic flux density is tesla
Magnetic flux $\phi = BA$
S.I unit of magnetic flux is weber



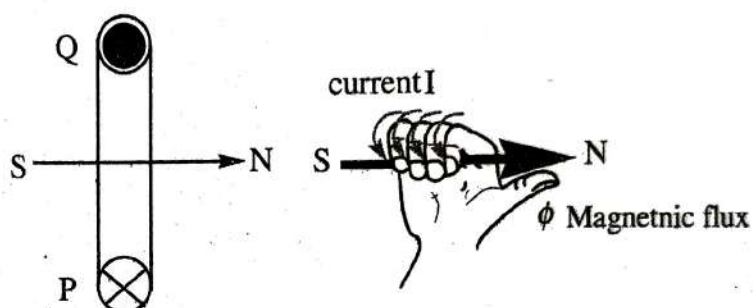
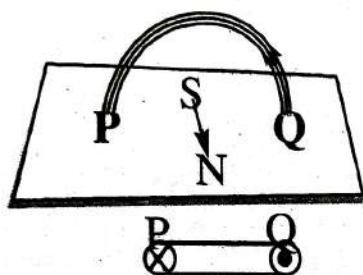
- Magnetic field is three dimensional around the sources such as bar magnet, conductor carrying current.
- The magnetic field at any point is characterised by the strength and direction.
- The strength of the field depends on the magnetic flux passing through given surface
- The direction of lines of force at a given point is denoted from $S \rightarrow N$ w.r.t compass
- The direction of lines of force is denoted from $N \rightarrow S$ w.r.t poles of magnet
- Magnetic lines of force are closed loops
- Magnetic lines of force do not intersect with each other
- Magnetic lines of force travelling in opposite direction repel each other

3.

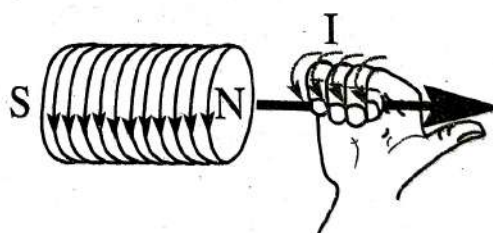
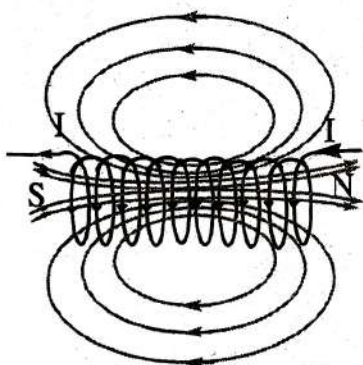
Straight wire carrying current



Circular coil carrying current



Solenoid carrying current



4.

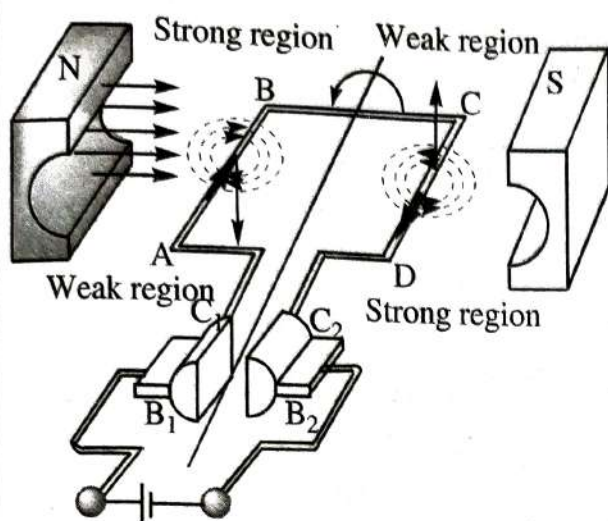
	A charged particle q entered with speed v into the magnetic field at right angles as shown below
	The magnitude of magnetic force resulted is $F = qvB$ The charged particle experiences some magnetic force in the direction which can be sought out by using right hand rule Here the magnetic force resulted $F \perp v$
	Radius of charged particle moving perpendicular in magnetic field $r = \frac{mv}{Bq}$ Time period of charged particle moving perpendicular in magnetic field $T = \frac{2\pi m}{Bq}$

5.

	Conductor of length (L) carrying current (I) Placed in a magnetic field (B) Experiences magnetic force (F) $F = ILB$ direction of force is given by right hand rule
Weak field	Magnetic force (F) is resulted on the conductor in the direction from strong field region to weak field region
Strong field	Repel each other Attract each other Weak field Strong field The magnetic lines of force due to applied field superpose with magnetic lines of force due to conductor carrying current As a result on the upper region we get weak field and in the lower region we get strong field

6.

ELECTRIC MOTOR



Converts electric energy into mechanical energy

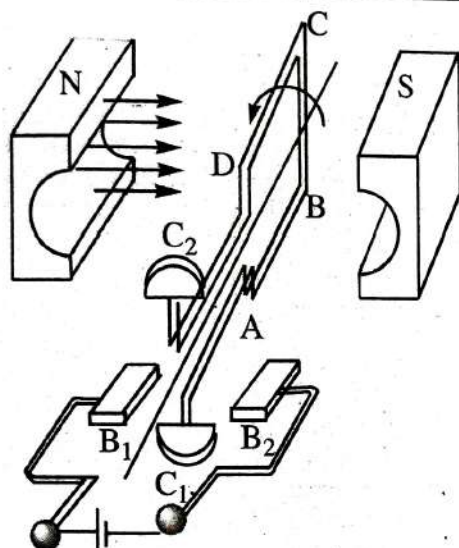
Works on basing electromagnetism

It contains rectangular coil carrying current

Placed in a magnetic field as we flow the current in the direction ABCD

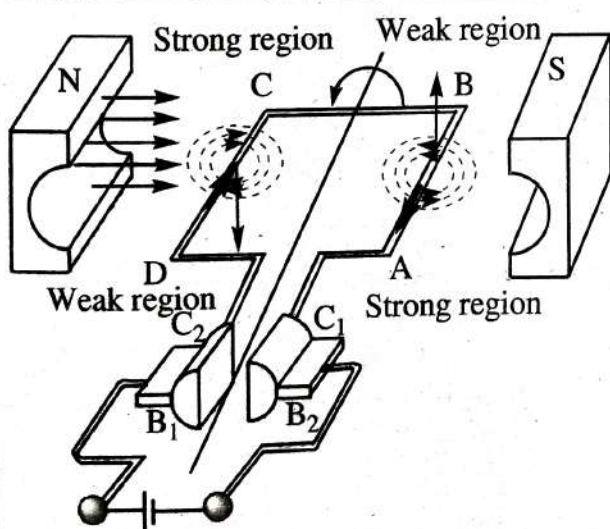
It experiences magnetic force such that on AB in downward direction and on CD upward direction

It starts rotating in anticlock wise direction



As it completes half revolution current is stopped because C_1 , C_2 lost their contact

C_1 , C_2 semi circular metal rings called commutator they help in changing the direction of current after every half revolution



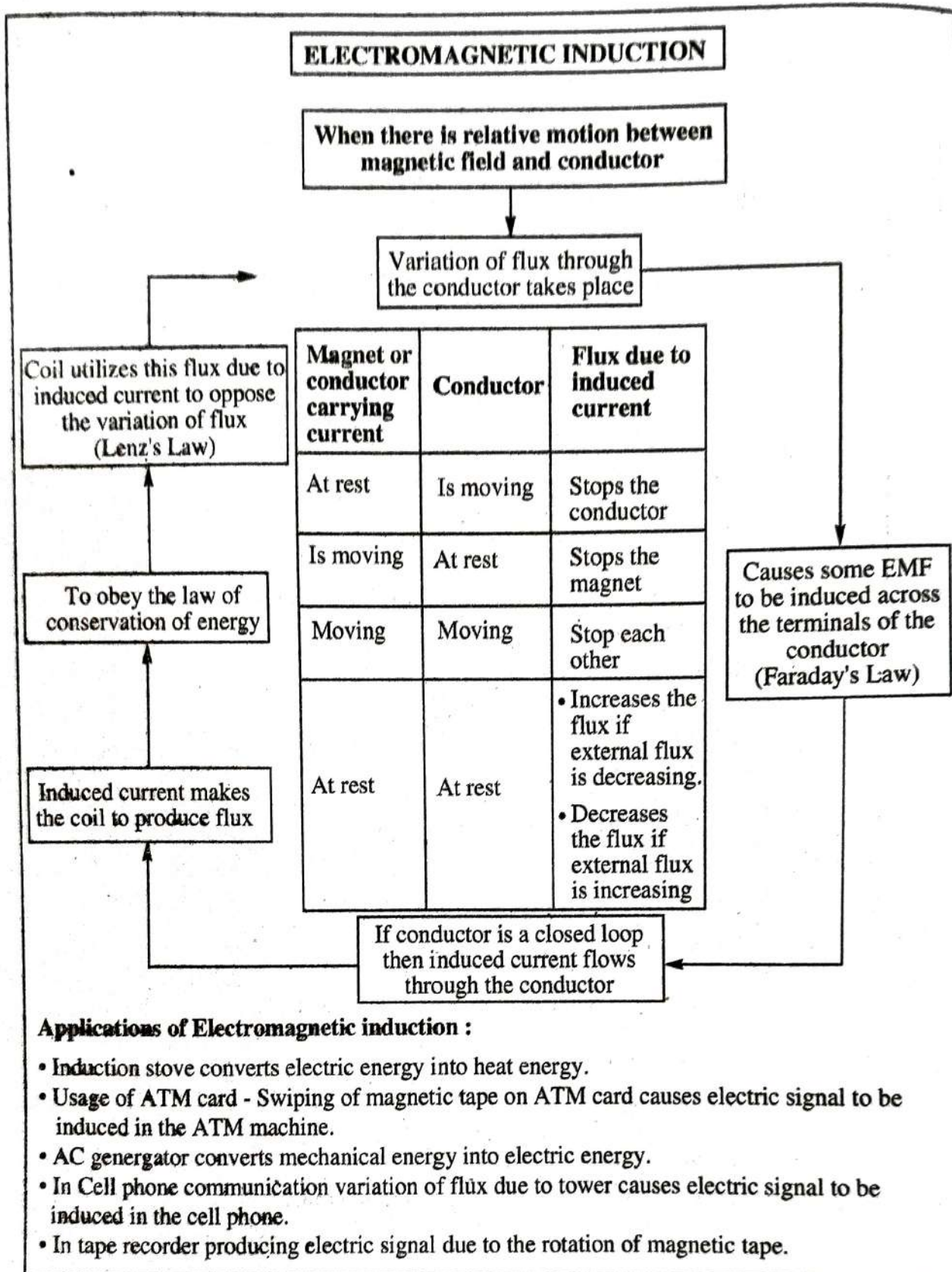
It continues revolution and again C_1 , C_2 will be in contact with brushes now current flows in the direction DCBA

It experiences magnetic force such that on DC in downward direction and on AB upward direction

Again rectangular coil starts rotating in anti clock wise direction and completes remaining half revolution

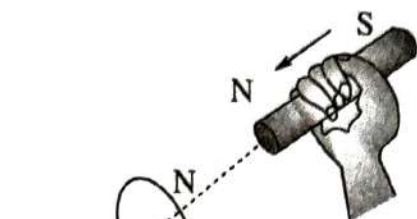
In this way it converts electric energy into mechanical energy

7.

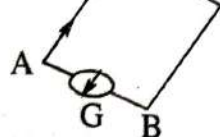


8.

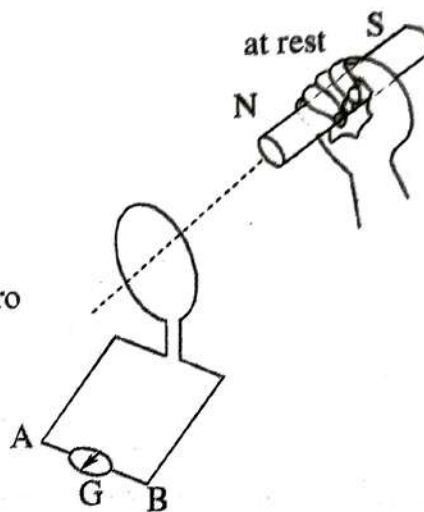
ILLUSTRATIONS OF ELECTROMAGNETIC INDUCTION



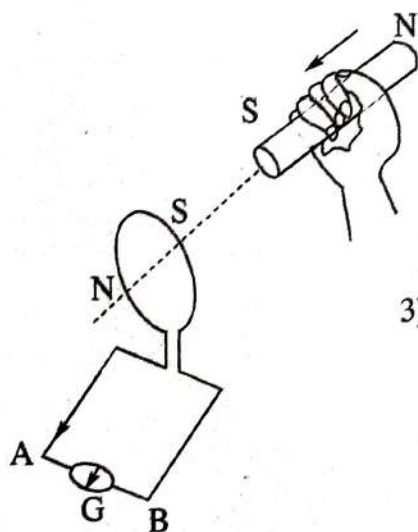
1) Coil repels the magnet developing North pole facing North by flowing current from B to A



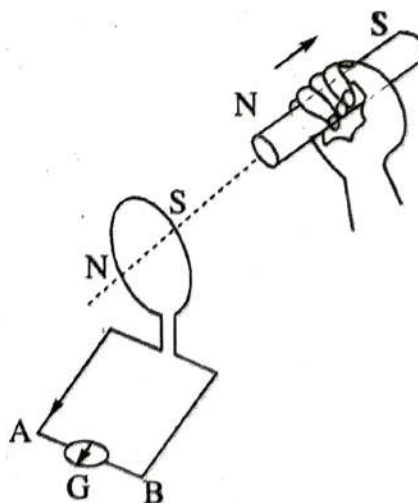
2) Coil neither attracts nor repels the magnet because induced current is zero



3) Coil repels the magnet by flowing current from A to B



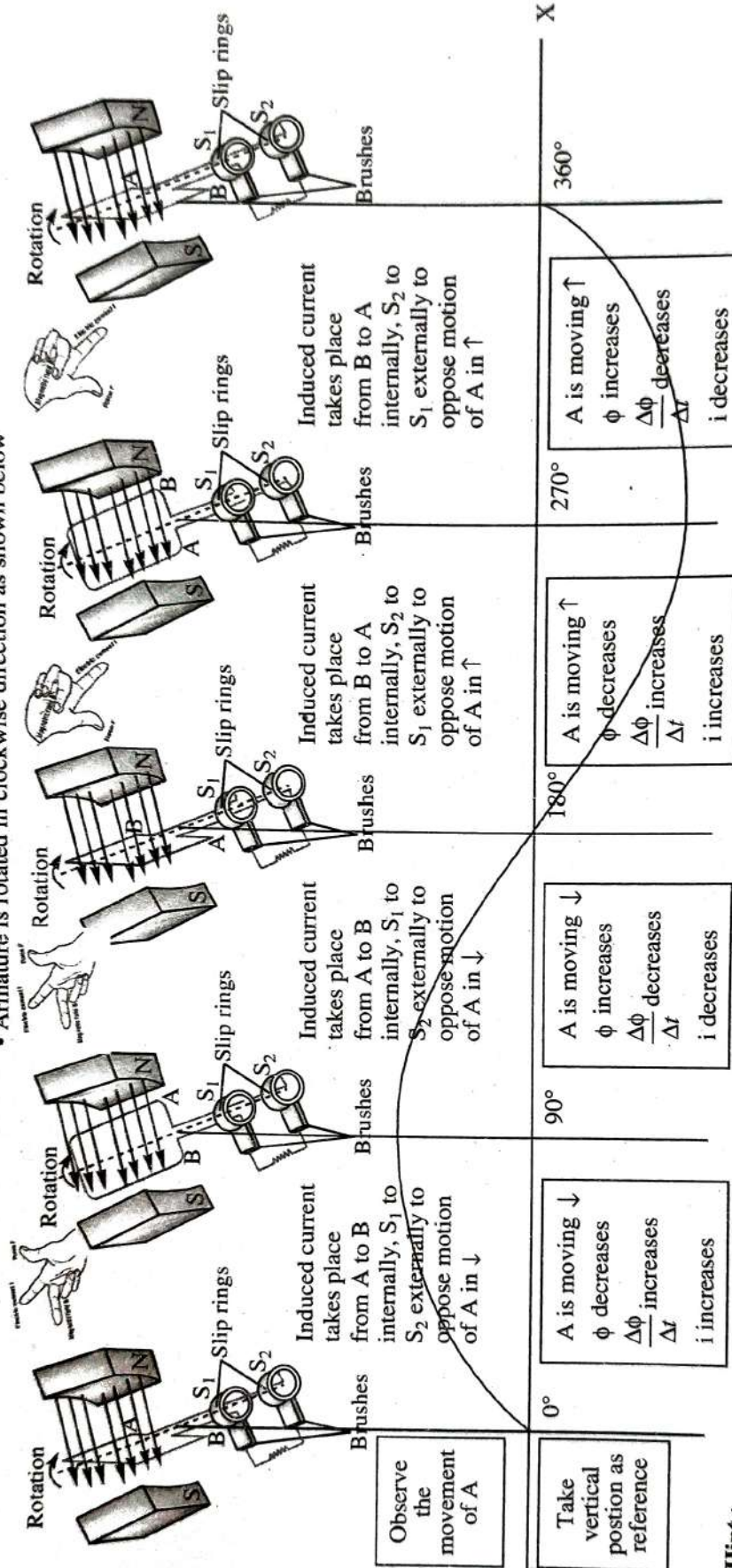
4) Coil attracts the magnet by flowing current from A to B



9.

AC DYNAMO (OR) AC GENERATOR :

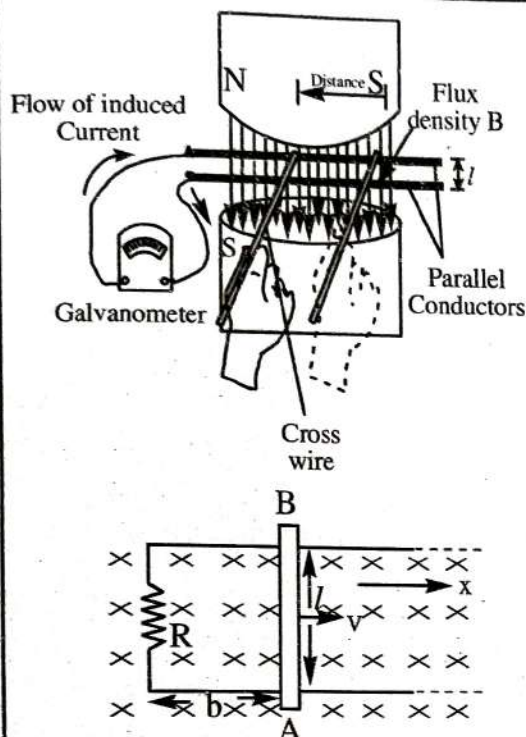
- It convert mechanical energy into electric energy.
- Works on basing electromagnetic induction.
- It contains rectangular coil placed inbetween permanent magnetic poles
- Terminals of the coil are connected to slip rings S_1 and S_2
- Armature is rotated in clockwise direction as shown below



10.

MOTIONAL EMF

In the given arrangement shown in the diagram if this cross wire is moved to the right, in contact with the two parallel wires as shown.



We get motional EMF across terminals of A, B due to electromagnetic induction,
 $\epsilon = Blv$

As we move AB to the right, the amount of magnetic flux enclosing the area

$$\Delta\phi = Blv\Delta t$$

By Faraday's law of EMI we get variation of flux ($\Delta\phi$) leads to induced EMF(ϵ)

The direction of induced current (i) has to take place in upward direction along A to B

The direction of force $F = ilB$, resulted is opposing the motion of the conductor (Lenz's Law)

The work done by the applied force is
 $W = ilBv\Delta t$

The work done by induced current
 $W = \epsilon i\Delta t$

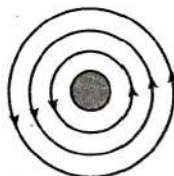
By law of conservation of energy we can have
 $\epsilon i\Delta t = ilBv\Delta t$

$$\therefore \epsilon = Blv$$

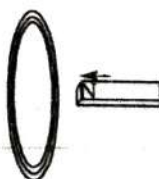
PRIORITY - I

Conceptual understanding (AS₁)

- Are the magnetic field lines closed? Explain, (T.Q) (Part-II.Pg.No. 110)
- See fig, magnetic lines are shown. What is the direction of the current flowing through the wire? (T.Q) (Part-II.Pg.No. 110)



- A bar magnet with North Pole facing towards a coil moves as shown in fig, What happens to the magnetic flux passing through the coil? (T.Q) (Part-II.Pg.No. 110)



4. A coil is kept perpendicular to the page. At P, current flows in to the page and at Q it comes out of the page as shown in fig. What is the direction of magnetic field due to the coil ?
(T.Q) (Part-II. Pg.No. 110)



5. The direction of current flowing in a coil is shown in figure. What type of magnetic pole is formed at the face that has flow of current as shown in figure ?

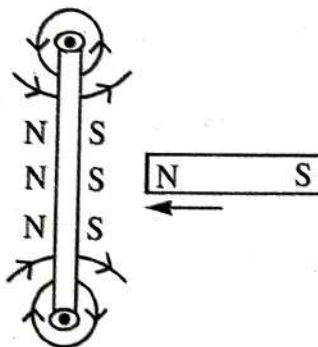


6. Explain Faraday's law of induction with the help of activity. (T.Q) (Part-II. Pg.No. 111)
7. What is the principle of an electric motor ? (T.Q) (Part-II. Pg.No. 113)
8. What is the role of the split ring in an electric motor ? (Part-II. Pg.No. 119)
9. State the rule to determine the direction of magnetic field around a current - carrying conductor. (Part-II. Pg.No. 120)
(S.S.C - 2015) (Part-II. Pg.No. 120)

(OR)

State the right hand thumb rule and state its significance ?

10. State the principle of an electric generator. (Part-II. Pg.No. 120)
11. Define the term "Induced current". (Part-II. Pg.No. 120)
12. State Faraday's law of electromagnetic induction? (Part-II. Pg.No. 120)
13. How do you determine the direction of induced current during electromagnetic induction ? (Part-II. Pg.No. 121)
14. State Lenz's Law ? (Part-II. Pg.No. 121)
15. List the properties of magnetic lines of force. (Part-II. Pg.No. 122)
16. Correct the diagram according to Lenz law and draw it again. SSC 2017 (Part-II. Pg.No. 122)



17. Explain the working of electric motor with a neat diagram. (S.S.C - 2018)(T.Q) (Part-II. Pg.No. 127)
18. What is the principle behind the working of induction stove ? Explain it and write its applications.

Asking questions and making hypothesis (AS₂)

1. Rajkumar said to you that the magnetic field lines are open and they start at north pole of bar magnet and end at south pole. What questions do you ask Rajkumar to correct him by saying "field lines are closed"?
(Part-II.Pg.No. 127)
2. What happens when current carrying wire placed in magnetic field ? (S.S.C - 2015)
(C.C.E) (Pg. No. 135)

Experimental skills and field investigation (AS₃)

1. How do you verify experimentally that the current carrying conductor experiences a force when it is kept in magnetic field ?
(S.S.C - 2018, 2019) (T.Q) (Part-II.Pg.No. 113)
2. How can you verify that a current carrying wire produces a magnetic field with the help of an experiment?
(S.S.C - 2016) (T.Q) (Part-II.Pg.No. 115)

Communication through drawing, diagram and graphs (AS₅)

1. Draw a neat diagram of an AC generator and name the parts.
(Part-II.Pg.No. 135)
2. Draw a neat diagram of electric motor. Name the parts.
(Part-II.Pg.No. 135)
3. Draw magnetic field lines around a bar magnet.

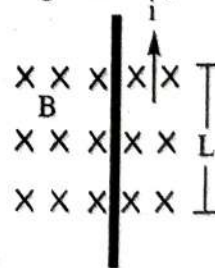
Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

1. How do you appreciate Faraday's law, which is the consequence of conservation of energy ?
(Part-II.Pg.No. 126)
2. Give a few applications of Faraday's law of induction in daily life. **SSC2024** (Part-II.Pg.No. 112)

PRIORITY - II

Conceptual understanding (AS₁)

1. Explain with the help of two activities that current carrying wire produces magnetic field.
(T.Q) (Part-II.Pg.No. 113)
2. Explain the working of AC electric generator with a neat diagram. (T.Q) (Part-II.Pg.No. 128)
3. The value of magnetic induction of uniform field is 2T. What is the flux passing through the surface of area 1.5 m² perpendicular to field ?
(T.Q) (Part-II.Pg.No. 126)
4. An 8 N force acts on a rectilinear conductors of 20 cm long placed perpendicular to magnetic field. Determine the magnetic field induction if the current in the conductor is 40 A.
(Part-II.Pg.No. 112)
5. Symbol 'X' indicates the direction of a magnetic field into the page.
A straight long wire carrying current along its length is kept perpendicular to the magnetic field. What is the magnitude of force experienced by the wire ? In what direction does it act ? (S.S.C - 2017) (T.Q) (Part-II.Pg.No. 111)
6. Define i) magnetic flux density ii) magnetic flux and give their expressions.
(Part-II. Pg.No. 153)



7. State the right hand thumb rule and write its significance in the case of
 - i) Straight wire carrying current
 - ii) Circular wire carrying current
 - iii) Solenoid carrying current

(C.C.E) (Pg. No. 135)
8. A charged particle 'q' is moving with a speed 'v' perpendicular to the magnetic field of induction B. Find the radius of the path and time period of the particle.

(Part-II. Pg.No. 109)
9. A coil of insulated copper wire is connected to a galvanometer. What will happen if a bar magnet is
 - i) pushed into the coil,
 - ii) Withdrawn from inside the coil,
 - iii) Held stationary inside the coil ?

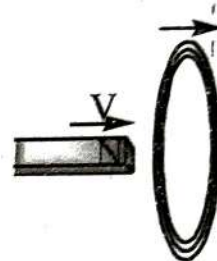
(Part-II. Pg.No. 124)
10. Explain the working of DC generator with a neat diagram

(Part-II. Pg.No. 129)
11. An alternating electric current has a frequency of 50 Hz. How many times does it change its direction in one second?

(Pg. No. 136)

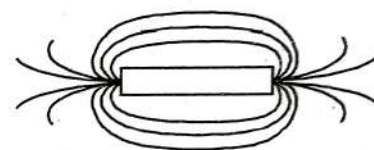
Asking questions and making hypothesis (AS₂)

1. As shown in figure both coil and bar magnet move in the same direction. Your friend is arguing that there is no change in flux. Do you agree with his statement ? If not what doubts do you have? Frame questions about the doubts you have regarding change in flux.



(Part-II.Pg.No. 112)

2. With the help of the figure given below the teacher explained that magnetic field lines are closed lines and not open lines. Write the questions which you will ask to test whether the given statement is right or wrong.



(C.C.E) (Pg.No. 136)

Experimental skills and field investigation (AS₃)

1. Describe an activity to demonstrate magnetic field due to straight wire carrying current ?

(Part - II Pg. No. 131)
2. Describe an activity to demonstrate magnetic field due to circular coil. (C.C.E) (Pg. No. 138)

Communication through drawing, diagram and graphs (AS₄)

1. Draw the neat diagrams of the following.

(C.C.E) (Pg. No. 137)

 - i) The magnetic field due to horse shoe magnet
 - ii) Magnetic field due to current carrying conductor.
 - iii) Magnetic field due to current carrying conductor placed in a magnetic field.

Aesthetic sense/ appreciation/ values (AS₆) / Application to daily life, concern to bio diversity (AS₇)

1. How do you appreciate the relation between magnetic field and electricity that changed the life style of mankind ?

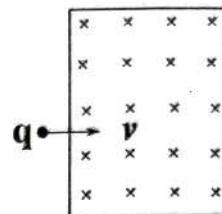
(Part-II.Pg.No. 126)
2. Which of the various methods of current generation protects nature well ? Give examples to support your answer.

(Part-II.Pg.No. 127)

PRIORITY - III

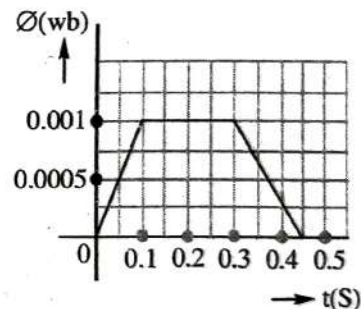
Conceptual understanding (AS₁)

1. A coil of area 2m^2 is placed perpendicular to the uniform magnetic field of 0.1 tesla. Find the magnetic flux linked with coil. If the coil is rotated by 90° in 2 seconds, how much e.m.f. is produced in the coil.
(C.C.E)(Part - II, Pg.No. 137)
2. A Magnetic field is acting perpendicularly into the plane of paper as shown in the diagram. A charged particle q entered in to the magnetic field with speed v at right angles to the direction of magnetic field then answer the following.
 - a) What is the direction of magnetic force resulted on the charge q
 - b) Calculate the magnitude of magnetic force resulted.
 - c) Calculate the workdone by the magnetic force on the charged particle ?(Part-II. Pg.No. 133)
3. Why the current carrying straight wire which is kept in a uniform magnetic field, perpendicularly to the direction of the field bends aside ? Explain this process with a diagram showing the direction of forces acting on the wire? (2017 SSC) (OR)



How do you explain the deflection of wire carrying current placed in a magnetic field.

4. The magnetic flux inside a coil of 400 turns changes for each single turn with time as shown in figure. Determine the maximum induced emf generated in the coil. Is there any change in induced EMF from $t = 0.1$ second to 0.3 second ? (TQ) (Part-II. Pg.No. 109)

Experimental skills and field investigation (AS₃)

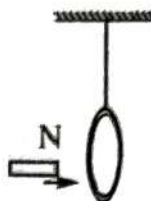
1. Describe an activity to demonstrate magnetic field due to solenoid. (Part-II. Pg.No. 132)
2. How do you demonstrate that the law of conservation of energy is applicable for electromagnetic induction. (M. Pg.No. 133)
3. What experiment do you suggest to understand Faraday's law ?
What items are required ? What suggestions do you give to get good results of the experiment ? Give precautions also. (T.Q) (Part-II. Pg.No. 114)

Information Skills (AS₄)Communication through drawing, diagram and graphs (AS₅)

1. Draw the diagrams of the following (C.C.E) (Pg. No. 139)
 - a) magnetic lines of force around a bar magnet
 - b) magnetic lines of force produced by a current carrying solenoid

Aesthetic sense/ appreciation/ values (AS_e) / Application to daily life, concern to bio diversity (AS_a)

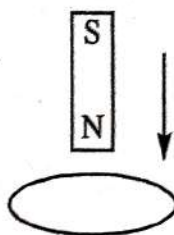
1. A circular coil is suspended from the rigid support as shown. A N - pole of a bar magnet is moved towards the coil perpendicular to the plane of the coil as shown (C.C.E) (Pg. No. 139)



- How do magnetic flux passing through the coil vary ?
- What is the direction of induced current in the coil with respect to direction of motion of bar magnet ?
- Draw the diagram of field lines at the plane of the coil due to bar magnet and also the field lines produced by the coil due to induced current in it
- Why current is induced in the coil ? Give reason.

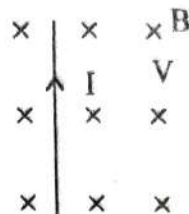
SESSIONWISE STUDENT TASK

- What type of energy conversion takes place in generators ?
- What are the devices which work on the magnetic effect of electric current ?
- What is the magnitude of force experienced by the wire if current carrying wire is kept parallel to the magnetic field ?
- What happens if slip rings are not there in the construction of Electric motor ?
- A bar magnet is moved towards the coil. Find the direction of the induced current and the pole formed on the top surface of the coil.



- What happens, if 'C₁, C₂' slip rings does not rotate along with armature ? Guess and write.
- An alternating electric current has a frequency of 100 Hz. How many times does it change its direction in one second?
- When an electron and a neutron are moving through uniform magnetic field with velocity 'v' perpendicular to the direction of magnetic field 'B'
 - Which particle deflects ? Why ?
 - What is the path of the deflected particle ?

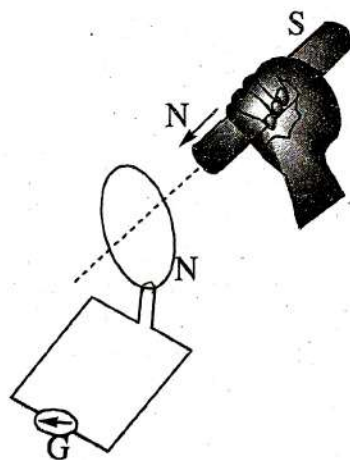
9. Observe the diagram. The magnetic field 'B' is perpendicular to the surface of the paper and current I is passing through the conduction wire.



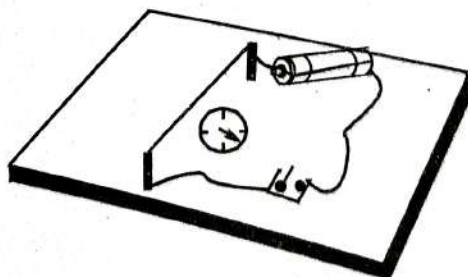
- How much of force is resulted due to the magnetic field ?
 - What is the direction of force resulted on the wire ?
10. A metal ring is inserted through a soft iron cylinder which is wound with a copper wire and fixed to the wooden base as shown in diagram.



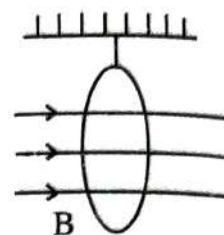
- When we flow an alternating current through the copper wire then what happens ?
 - How do metal ring behaves as a magnet ?
 - What is responsible for the current in the metal ring ?
 - What do you name this phenomenon ?
 - How do you describe this phenomenon ?
11. In an experiment
- What are changes that occur when a bar magnet is pushed and pulled quickly by Sriharsha in to the given circuit?



- What are the changes that occur in the directions of compass in the diagram when the direction of current is changed in a circuit?

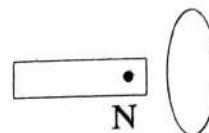


12. Observe the diagram. A loop was hanged as shown in the figure and its cross sectional area is 'A'. The magnetic flux density at the face of the loop is 'B'. Then what is the flux related to the loop. If the loop is rotated with the base of hanging to rope, then is there any chance of change in the magnetic flux? Explain?



13. Observe the diagram. A bar magnet and a loop are given. Find how the flux is changing in the following situations with respect to loop and explain the reasons.

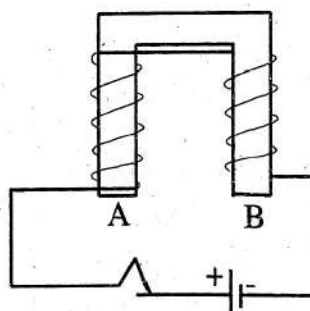
- The loop and bar magnet are at rest.
- Loop is at rest but magnet is moving towards loop.
- bar magnet is at rest, but loop moves away from the bar magnet.
- If loop and bar magnet are having same velocity.



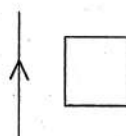
14. Saritha read in a book that the voltage induced across the terminals of a coil does not depend on its resistance. How do you analyse this information is correct or not? How do you prove it?

15. The diagram shows a coil wound around a U shape soft iron bar AB.

- What is the polarity induced at the ends A and B when the switch is pressed?
- Suggest one way to strengthen the magnetic field in the electromagnet.
- What will be the polarities at A & B if the direction of current is reversed in the circuit?



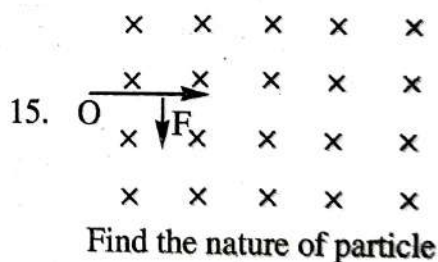
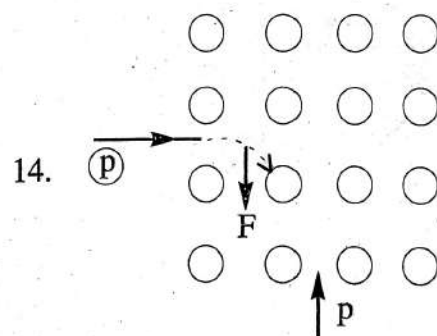
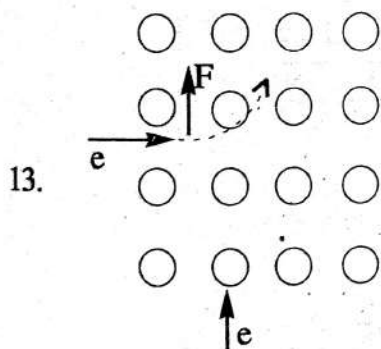
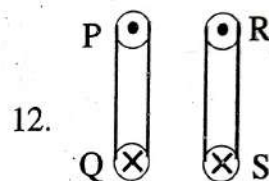
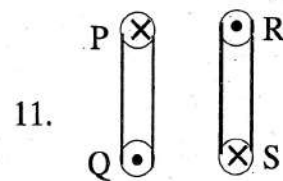
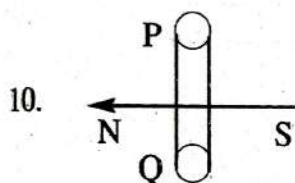
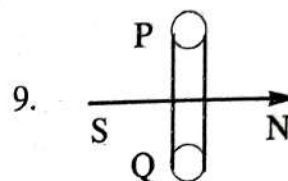
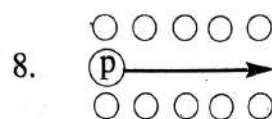
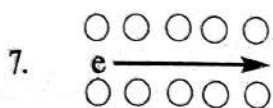
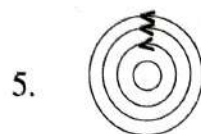
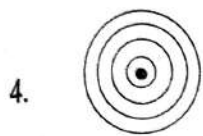
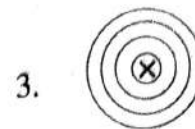
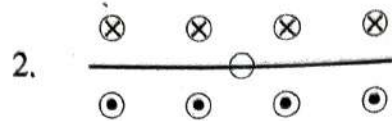
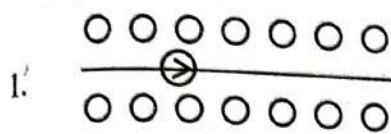
16. Current carrying conductor is present near a square loop as shown in figure.



If current through the conductor is stopped suddenly, mark the direction in which current is induced in the square loop.

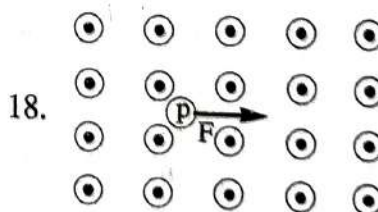
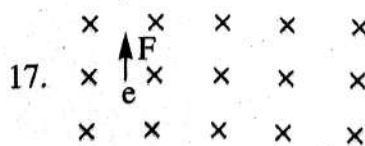
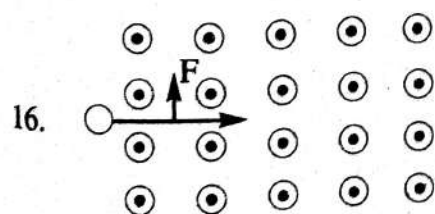
- Even if the net force acting on the coil is zero placed in a magnetic field, the coil still rotates, Why? Imagine & Write.
- "Current carrying conductor behaves as a source to generate magnetic field" Write the apparatus and explain its experimental procedure?

ELECTROMAGNETISM PRACTICE DIAGRAMS



Find force direction

Find force direction



Find the nature of particle

19.



20.



21.



22.



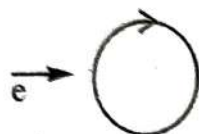
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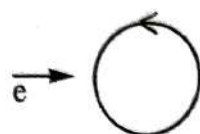
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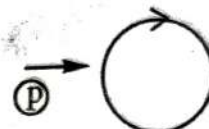
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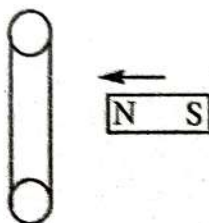
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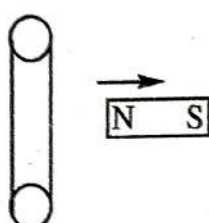
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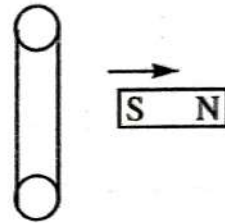
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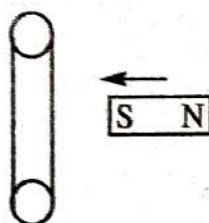
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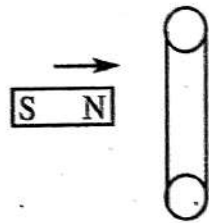
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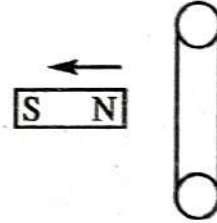
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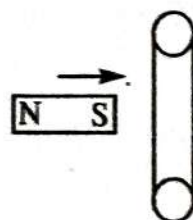
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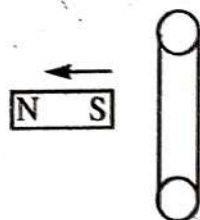
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34.



35.



36.

